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BACHELORTHESIS

Analyzing and predicting user behaviour in online marketing

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Formal declaration

I declare on oath that I have written this bachelor's thesis independently and exclusively using the literature and other resources provided. The thesis has not been submitted in the same or a similar form to any other examination authority for the award of an academic degree.

Karlsruhe, 29.02.2024

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List of Abbreviations

List of Abbreviations

AD	Advertisement
API	Application Programming Interface
BAS	Behavioural Activation System
BERT	Bidirectional Encoder Representations from Transformers
BIS	Behavioural Inhibition System
CNN	Convolutional Neural Network
CTR	Click-Through-Rate
CV	Computer Vision
GB	Gradient Boosting
GPT	Generative Pre-Training Transformer
LLM	Large Language Model
NLP	Natural Language Processing
NN	Neural Network
NRC	National Resource Council
PCC	Passively Consumed Content
RF	Random Forest
SIG	Semantic Information Gain
SVM	Support Vector Machine
TF-IDF	Term Frequency – Inverse Document Frequency
UGC	User Generated Content
VGG	Visual Geometry Group

1. Introduction

The scope of this present thesis is to identify, extract and evaluate variables that carry predictive potency in forecasting customer behaviour in online marketing. These variables are extracted solely from contextual information present in or surrounding placed advertisements and their predictive capabilities need to be evaluated on real-world customer interactions.

1.1. Motivation

In recent years, a steady increase in global spendings on advertisement is observable that is prognosed to culminate in a duplication of spendings within one decade, starting at 650-billion dollars in 2018 and reaching an estimated all time high of 1.300-billion dollars in 2028 (Statista, 2023). Not least caused by the global corona pandemic in the early 2020s that shifted public and corporate life from the public to the virtual abruptly, the ongoing digitalization of our society has led to a substantial growth in digital marketing budgets. As traditional advertisement media is predicted to fall by over 20% in the next years (FinancesOnline, 2024), these freed capacities are taken over by the digital market.

While traditional advertisement revolves around addressing wide ranges of potential customers by trying to appeal to a cross-section of society, the digitized marketplace allows for a highly individualized targeting, aiming for an optimal advertisement-person fit (Strauss et al., 2005). Therefore, gathering customer information has quickly become the focal point for advertisers, resulting in data being treated as its own currency. Most approaches rely on so-called 3rd party cookies (cookies that are placed by a domain or in other words a party that is different to what is shown in the browser's address bar) to track user journeys and habits when browsing the internet to gain insights on predictable customer behaviour (Cahn et al., 2016). This information is then used to provide highly adapted and individualized advertisements based on the user's inferred preferences.

With the advent of increasingly strict laws to enforce data privacy and regulation such as the European Digital Services Act (European Commission, 2024), alternatives to the lucrative usage of 3rd party cookies to deduce user preferences are of need and are likely to prosper in the next years. Consequently, approaches that identify user preference and behaviour without relying on personal and identifiable data are on the rise. A promising approach is to leverage contextual information that is accompanied by the advertisement itself, based on the assumption

Problem Statement and Setting

that a user expressed his interest in said contextual information by voluntarily choosing to interact with it. In other words, a user that reads an article about the latest football results is likely to be interest in football.

1.2. Problem Statement and Setting

To its core, (online) marketing is faced with the challenge of identifying and addressing user preferences to tailor advertisements accordingly. Since user preference by itself is a latent construct that is hard to measure, it is commonly approximated by user behaviour, reasoning that preferences are expressed through user behaviour (Otamendi & Martín, 2020). For the online environment, user behaviour can be measured empirically in form of attention time given to an advertisement or in the form of clicks onto said advertisement. Consequently, identifying extractable constructs that have a direct impact on human behaviour are required.

For this present work, the following setting is given as a set of fixed requirements that need to be considered when trying to predict costumer behaviour:

As a first requirement, the available information is restricted to online advertisement of webpages and the available information is restricted to the:

- *Context*, defined as an article provided by a publisher that the advertisement appears in. Contexts are mainly composed of text but often also include images.
- *Advertisement*, defined as the object of interest for which user interaction is intended to be maximized for. Advertisements also consist of text and images.

Additionally, construct candidates need to be feasible and measurable within the online environment. As an example, measuring user interest by utilizing biofeedback might produce good results in predicting user behaviour but is not feasible for an online environment.

Lastly, both advertisements and contexts are only available in German and are passively consumed content (PCC), which refers to the fact that the contents are read or processed by the user instead of being written or produced by themselves, which will consequently be named user generated content (UGC). The differentiation and consideration of these two types of content is pivotal, as any chosen construct is supposed to evoke its effect via its passive consumption (PCC) rather than as a direct expression of author intention (UGC). To illustrate the relevance for the given setting, the following example is given: Using twitter posts to extract a specific construct would not be suitable for the given setting, as any extracted information

Structure of this thesis

will most likely express intentions of the tweet's author due to its nature as UGC. On the other hand, leveraging journalistic content is more suiting, as the journalist wrote the article as a service for his potential reader, utilizing rhetorical elements to catch his audience's interests. In summary, UGC can be considered author-centered and PCC reader-centered.

To solve the presented problem, the following tasks need to be processed:

- Identification of the most promising construct to predict user behaviour,
- identification of the most promising extraction modality (text, image, both),
- construction of a German dataset,
- extraction of the most promising feature sets,
- training and evaluation of classification algorithms,
- performance evaluation in predicting user behaviour on real-world data.

1.3. Structure of this thesis

To identify best constructs for user behaviour prediction, this work will provide a thorough examination and comparison of construct candidates in chapter 2 [Fundamentals and related work] and deduce an optimal construct based on the provided requirements and the theoretical insights from the fields of psychology, sociology and neurophysiology. Second, recent work on text and image processing is evaluated under consideration of the described properties for both context and advertisement and an approach for consumer behaviour prediction is proposed, including data set, feature set and construct taxonomy configurations. Chapter 3 [Design and Implementation] describes the implementation of the proposed approach, focusing on algorithm optimization for the first part and evaluating the models' effectiveness in predicting user behaviour on real-world data in the second part. Chapter 4 [Results] provides insights on the performance of the chosen approach by evaluating and comparing experimental results. A final summary and review on the executed experiments is given in chapter 5 [Discussion] by drawing a comprehensive conclusion, outlining limitations of this present work and proposing possible next steps for future research.

2. Fundamentals and Related Work

Human behaviour and decision making are complex by their nature and are subject to many prominent research areas such as psychology, sociology and neurophysiology. To predict human behaviour in an online marketing environment, one must first understand the underlying mechanisms that influence behaviour and decision making, which will be dealt with in the upcoming chapters.

2.1. Ad-Context Congruency and Interaction Effects

To understand human behaviour in the outlined system's setting, namely consisting of a contextual and an advertorial content, the interplay and interactional patterns between the two aspects will be considered first.

The research field of congruency deals with this subject, analyzing the degree of similarity between an advertisement and the media content the advertisement is placed in. As underlying mechanisms, Contextual Priming and Fluency are commonly referred to (Segev et al., 2021).

The Contextual Priming Effect describes the activation of a series of related concepts in consumers' semantic networks when processing a target. For targets relevant to the known concepts, or in other words congruent with the concept, their semantic processing becomes easier which in turn affect consumers' subsequent evaluation of the target (Higgins, Bargh, & Lombardi, 1985). In advertisement, the ad itself serves as the cognitive prime that activates consumers' knowledge concepts. In the case of ad-context congruency, an association with the activated concept is more likely, leading to more favourable evaluations compared to incongruent ad-context pairings (Janssens, De Pelsmacker & Geuens, 2012).

The Processing Fluency Model claims that for consumers undergoing prior exposure to a stimulus, ensuing processing is perceived to be more fluent which in turn leads to more favourable attitudes towards the target (Seamon et al., 1995). For the holistic Fluency model, a differentiation into Perceptual and Contextual Fluency is proposed. Perceptual Fluency focuses on the fluency experience concerning shape, size or colour based on previous exposure, whereas Contextual Fluency is related to cognitive evaluation and attitude toward the target. Therefore, Contextual Fluency captures the ease of association an individual experiences when relating to and processing a target it was previously exposed to (Tversky & Kahneman, 1973).

The importance of congruency in consumer behaviour is stated by Germelmann et al. (2020), showing that thematic ad-medium congruency has a positive effect on consumer evaluations of the advertised product, whereas thematic incongruency results in the opposite. Besides thematic congruency, design congruency is also explored, which relates closely to the concept of Perceptual Fluency. Bishop & Brocato (2015) could show that style congruency, differentiating highly visual and highly informational content, increases advertisement success.

Furthermore, congruency in more latent and abstract concepts such as mood have also been explored. Noland & Wen (2022) demonstrated that consumers' mood interacts with advertisement design when it comes to advertisement perception. According to their findings, test subjects in a negative mood reacted more positively to advertisement presented in a structured and information-oriented way, whereas subjects in a positive mood preferred advertisement in a narrative style. Similar effects were found by Gorn in 1982, revealing that hearing liked or disliked music while being exposed to an advertised product directly influences subsequent product choice.

In summary, when trying to predict advertisement success, a holistic consideration of both the advertisement and its surrounding context is important. The concept of congruence demonstrates that interactions between the two elements exist, and that the consumer can be affected and primed by contextual information. Besides basic congruency concerning design and topic, it was also shown that latent constructs such as mood can influence the perception and appraisal of the presented advertisement. Therefore, identifying such latent constructs that evoke behavioural patterns is important for the prediction of user behaviour in online marketing.

Departing from the interactional examination, the upcoming segment will discuss possible behavioural taxonomies that can be extracted for both context and advertisement.

2.2. User decision and behaviour in online marketing

To design a system that predicts user behaviour, one must first consider general human decision theory. For media and online marketing, stimulus-response models that originated from behaviouristic psychology are commonly used to describe decision processes (Jacoby, 2002). In its core, the theory postulates that consumers respond in a foreseeable manner to marketing stimuli such as advertisement content. Nowadays, the theory is outdated in media research due to its lack of consideration of an active consumer role in the process of decision making. It is replaced by humanistically driven concepts such as the Uses and Gratifications Approach

proposed by sociologist Elihu Katz (Katz et al., 1973) that puts stronger emphasis on the individual's desires and interests at decision time, thereby assigning an active decision role to the individual to overcome the strict "black box" thinking that is predominant in classical behaviourism and introducing more complexity to decision processes. Still, the core principle of foreseeable reactions to stimuli that can be designed and manipulated by content and media creators introduced by the stimulus-response model holds true for any modern model.

Building on this basic principle, a range of theories intent to identify its underlying patterns. The Behavioural Inhibition System (BIS) and Behavioural Activation System (BAS) or Fight-Flight-Freeze System, developed by biopsychologist Alan Gray (Gray, 1970) is one such example. The theory differentiates two systems that regulate avoidance and punishment motivation on the one and reward and approach motivation on the other side. Thereby, the theory states distinct reactional patterns as responses to environmental stimuli depending on the behavioural system that is being triggered by the situational context. In short, people can either be attracted by a stimulus when interacting with it, or not. This becomes more evident when taking its alternative name into consideration: Fight, Flight & Freeze define the three evolutionary and basic responses to environmental stimuli that require interaction. The fact that researchers have found strong relations between an individuals' BIS and BAS system and their consumer-related activities (Davidson & Irwin, 1999) stresses the concept's relevancy for marketing applications. Neurophysiological studies have shown that when a reward is claimed, brain activity in regions that are typically associated with the BAS system correlates with subjective reports on the experience's pleasantness.

Departing from mere evolutionary or behaviouristic models, the Cognitive Appraisal Theory takes the cognitive processing and evaluation of an incoming stimulus into account and claims that subjective stimulus interpretation plays a crucial part in reaction elicitation. Most notably used by Lazarus and Folkman (1984) in their transactional model of stress and coping, Cognitive Appraisal is another example of stimulus-reaction patterns and is used to explain a range of constructs such as stress, mental health, coping or emotion.

One of the most influential contributions of recent times to the field of decision theory is the work of Daniel Kahneman (Kahneman & Egan, 2011), postulating two separate systems in processing incoming stimuli. As one of the most popular overarching frameworks in describing the human brain's working mechanisms, it puts emphasis on the importance of conscious and unconscious processes in decision making. System one is fast, automatic, and not under

volitional control whereas system two processes the opposite: slow, voluntary, and controlled decisions.

Applying this Kahneman's model to marketing, it becomes evident that both systems require attention when user behaviour is supposed to be predicted. Besides addressing the rational and slow system two, the irrationality underlying system one also plays a vital role in the emergence and understanding of consumer behaviour (Bechara & Damasio, 2005). To stress the relevancy of this theory and especially the increasing consideration of system two in marketing, Escalas (2007) showed that consumers are more likely to be persuaded by affective and thereby predominantly unconscious outcomes rather than by strong logic arguments present in information given by an advertisement.

In summary, consumer behaviour and decision can be modelled via a stimulus reaction scheme that allows to predict outcomes of presented stimuli. The presented theories differ in their reasoning for how and why a stimulus provokes a reaction. Whilst BIS & BAS approaches focus on the evolutionary aspects for stimulus response patterns, cognitive appraisal theories depart from the narrow and instinct based concepts, as they include cognitive evaluation and appraisal of a stimulus into the process of decision making and stimulus response. Lastly, Kahneman's model of two distinct behavioural systems introduces the concept of conscious and unconscious reactional patterns. The uniting theme over the mentioned approaches is the idea of predictable and foreseeable behavioural and reactional patterns to stimuli the individual is exposed to.

As shown in the section above, both conscious (following the cognitive appraisal theory that requires active stimuli processing) and unconscious (evolutionary reactions to stimuli based on the BIS and BAS system or Kahneman's system two) processes can lead to predictable reaction patterns. Following this line of thought, the construction of a system to automatically predict user behaviour is reasonable and needs to focus on identifying the most suitable stimulus constructs to predict the desired user behaviour, or in other words a purchase intention, by considering both conscious and unconscious processes.

In the next section, potential stimulus constructs are presented and analysed in relation to their potential for predicting consumer behaviour and decision.

2.3. Comparison of stimulus constructs

Since human behaviour is complex by nature, enlisting all variables that theoretically impact the decision process is impossible and by no means within the scope of this work. Nevertheless, some limitations were used to focus on relevant variables.

First, some constructs have been proven to influence decision processes but are primarily linked to the personality of the consumer rather than the context he or she is being exposed to. Examples for this would be personality traits such as extraversion, openness, narcissism and so on. As an example, Slater and Rouner (2002) found that individuals with a high sensation-seeking personality, which is highly correlated to extraversion, are more receptive and responsive to messages high in narrative elements such as drama, emotion and event. General sociodemographic variables such as age, gender or political orientation are other examples.

Second, constructs may not be possible or highly infeasible to measure given the described requirements. This also includes constructs of the first category, since information such as gender, age or political orientation cannot be accessed directly for the given *cookieless* environment.

In addition to this category, some other constructs such as text engagement, immersion and flow experience are expected to influence decision processes. These related concepts describe the interaction intensity an individual perceives while interacting with a presented content. Whilst engagement is often defined as the “emotional, cognitive and behavioural connection that exists between a user and a resource” (Attfield et al., 2011), immersion and flow are described as positive mental states of optimal experience arising from intense involvement in an activity that is enjoyable (APA Dictionary of Psychology). For the relevant online marketing environment, Chen & Yao (2022) identified that a media experience high in narration and immersion strengthens the individuals’ positive attitude towards the general experience and thereby affects its persuasive outcome, leading to a higher purchase intention.

Even though being relevant and interesting concepts that should be considered potential candidates for the reasons given above, they cannot be measured in the given online marketing environment, as neurophysiological markers such as gaze and eye tracking would be needed for proper assessment (Cherubino et al., 2019). To handle the measurement and extraction problem for online advertisement, some approximation methods have been proposed.

Most of these approximations are based on the related metric of dwell time, which is the time a user spends on certain segments of a page or its content. A Google research team led by Lagun

Comparison of stimulus constructs

and Lalmas proposed to leverage viewport information, defined as the “position of the webpage portion visible to the user at any time” (Lagun & Lalmas, 2016). Using the viewport information, dwell time distributions for any webpage can be computed and used to cluster differing engagement classes.

For example, the authors differentiate bounce, shallow, deep and complete engagement. Bounce describes the minimal engagement class where users directly leave the page after opening it, whereas complete on the contrary is a full read of the webpage, meaning the user scrolled down to the bottom of the page.

Related to the dwell time approximation, Grinberg proposed the concept of Semantic Information Gain (SIG) in 2018. In short, SIG computes high dimensional textual embeddings of the whole text as well as embeddings for each paragraph and computes cosine similarities between each paragraph and the entire text. Thereby, the semantic richness for each paragraph is derived which, like Lagun’s and Lalmas’ viewport approach, also results in a information distribution over the whole text which can be clustered into different engagement classes.

Even though these approximations seem promising, they contain problems that must be addressed. Being approximations, they are prone to measurement inaccuracies and errors. For the viewport approach, problems arise when users are inactive for a longer period. This may be caused by leaving sessions open, using second screens or simply by leaving the computer with the viewport fixed at a certain position. This issue does not arise for the SIG approach since it does not measure real-time metrics and relies solely on textual information. However, this reliance on textual data raises questions about the actual approximation quality of the approach. The approach relies on the assumption that semantically dense text segments correlate with high text immersion, which cannot be validated and therefore leads to the introduction of bias and error proneness. Therefore, the approximation approaches are excluded from consideration.

Beyond those limitations, the three constructs Persuasion, Sentiment and Emotion were identified as most promising and are evaluated.

2.3.1. Persuasion

Persuasion can be described as the “process by which a communicator attempts to convince a person or group to change their attitude or behaviour regarding a particular issue” (Perloff & Richard, 2010) and has captured researchers’ interest since the advent of mass media. Persuasive messages can use different forms of stimuli like text, image or sound and aim to alter behaviour or choice by systematically applying persuasive strategies.

One of the most popular taxonomies for persuasive strategies was introduced by social psychologist Robert Cialdini, differentiating six strategies: Reciprocity, Scarcity, Commitment, Consensus, Liking and Authority (Cialdini, 1984). Each of these strategies define predictable behavioural patterns that can be evoked by presenting a corresponding stimulus. As an example, a typical scarcity stimulus is to stress the limited product availability or time boxed offerings in advertorial messages. Persuasive messages based on reciprocity focus on the benefits and services or favours the vendor offers in return for a purchase, like prolonged warranties, special price reductions and so forth. Being a well-studied field in social sciences, there is a large body of research identifying and underlining the presence and effect of various persuasion strategies (Kumar et al. 2023).

Recent work of Kumar et al. (2023) builds on these principles and develops a more fine-grained taxonomy for persuasive strategies, consisting of nine persuasion strategy groups that can be further divided into 20 actual strategies. The research group provides a first attempt to bridge the gap between social science and computational modelling of persuasion by publishing a dataset of labelled advertisement images as well as a multi-modal classification algorithm to predict persuasive strategies based on the provided dataset. The dataset builds on the Pitt Image Ads dataset (University of Pittsburgh, 2018), that consists of over 60.000 poster advertisements and enriches part of it with its persuasive strategy. The proposed algorithm combines both textual features extracted via an optical character module from the poster and image features in an attention fusion model to predict a dominant persuasion strategy.

In summary, Persuasion by its definition is a useful concept to predict user behaviour, as the strategies are designed to manipulate and alter customer decision and behaviour in line with the interests of the advertiser. For the present requirement setting, some problems arise when including Persuasion for predictive purposes:

First, even though first datasets are becoming available like the one provided by Kumar et al., dataset scarcity is still a challenge. Since advertisement design can differ quite substantially,

the generalization capabilities of a classifier trained on the provided dataset are highly questionable, as the advertisements present in the dataset are solely posters that put a strong emphasis on image features. The required advertisement design described in the requirements in chapter 1.2. on the other side combines text and image features, which might lead to problems. Second, Persuasion as a construct is by its definition limited and tailored to advertisement contents. Even though the application of a learned persuasion representation to the context of the advertisement is technically possible, it is unlikely to yield interpretable results, being composed of newspaper articles. In journalistic writing, the application and usage of persuasive principles cannot be expected to occur as a rhetoric device.

2.3.2. Sentiment

Sentiment is defined as “an acquired and relatively permanent major neuropsychic disposition to react emotionally, cognitively and conatively toward a certain object (or situation) in a certain stable fashion, with awareness of the object and the manner of reacting” by Cattell (2006). Sentiments are formed when a certain object is actively processed by a person for a longer period, resulting in a dispositional idea towards the object. This idea involves predictable emotional tendencies, being evoked whenever a subject gets in contact with said object (Munezero et al., 2014). Following this definition, sentiments are enduring mental attitudes that provoke an emotional reaction, therefore being of interest for this work. According to this description, concepts such as loyalty, friendship, love or patriotism can be assigned to the sentiment class.

In practice though, research on sentiment extraction almost exclusively focusses on the reduction to a positive and negative emotional disposition transmitted via the content (Kheiri & Karimi, 2023; Li et al., 2022), with some approaches including a third neutral category.

Sentiment analysis, especially in Natural Language Processing (NLP), is a well-researched field, leading to a large body of available extraction algorithms and datasets that are capable of extracting sentiment tendencies well. On the other hand, the reduction to a maximum of three classes decreases the concept’s potential to differentiate user behaviour considerably. Also, being an enduring rather than a short mental state, the question arises whether the reaction intensity evoked by sentiments is high enough to justify its further examination for the purpose of this work. Dealing with passively consumed content, it is important to choose a construct that has the highest potential to be transmitted on to the reader after stimulus exposure. As sentiments are stable by their definition, it seems implausible to assume such a transfer process

for PCC. Furthermore, even though the definition posits distinct behavioural reactions to sentiments, these reactions are described as emotional tendencies. Therefore, a direct focus on emotion extraction seems more plausible.

2.3.3. Emotion

The concepts of emotion and sentiment are often used interchangeably. This is mostly caused by the fact that both refer to “experiences that result from the combined influences of the biological, the cognitive and the social” (Stets, 2003). In addition, the last segment demonstrated the entanglement of the two constructs, as sentiments are related to emotional reactions. When reducing sentiments to positive and negative, emotions can also be segregated into predominantly negative (e.g. *fear*, *sadness*) or positive (e.g. *joy*, *excitement*) demonstrating the further entanglement. For differentiation of the concepts, overall state endurance is used most commonly. Whilst emotions are brief episodes of brain, autonomic and behavioural change, sentiments have been found to form and be held for a longer period (Munezero et al., 2014).

When searching for a holistic definition of emotions, Kleinginna and Kleinginna (1981) provide an all-embracing description for the construct: “Emotion is a complex set of interactions among subjective and objective factors, mediated by neural/hormonal systems, which can a) give rise to affective feelings of arousal, pleasure/displeasure; b) generate cognitive processes such as emotionally relevant perceptual effects, appraisals, labelling processes; c) activate widespread physiological adjustments to the arousing conditions; and d) lead to behaviour that is often, but not always, expressive, goal-directed, and adaptive”. This definition provides both explanations for emotion causes and effects.

When examining emotion causes, emphasis is put on neural/hormonal systems. This is in line with later work by Damasio et al. (1991), stating the relevance of emotions for rational thinking and behaviour by developing the Somatic Marker Theory.

Somatic makers are brain structures located in the ventromedial prefrontal cortex that can either be genetically inherited or emerge due to learning and experience. Thereby, somatic markers provide a neural representation of experienced emotions and their associated reaction and decision patterns, underlining the pivotal importance of emotions for human behaviour and decision-making. From an evolutionary perspective, this mediation effect of emotions in the decision process by relying on experience increases decision speed compared to thorough and cognition-intense evaluation processes.

Comparison of stimulus constructs

The relevance of the evolutionary aspect of emotions is underlined by evolutionary psychology, postulating that different emotions emerged as responses to specific adaptive problems that were relevant in the ancestral environment (Cosmides & Tooby, 2000) and thereby formulating clear distinctive action tendencies. As an example, experiencing fear can lead to self-protective mechanisms to avoid harm which increases chances of survival (Griskevicius et al., 2009). Anger on the other hand typically results in action tendencies that stress one's body size, talking louder, fighting or allocating cognitive resources to outcompete potential opposing agents (Frijda, 1986).

So far, the described cause patterns rely on spontaneous and largely uncontrollable mechanisms, mainly characterized by a lack of required cognitive involvement such as thinking, reasoning or consciousness. Nevertheless, cognition can play an important role in the genesis of some emotions. The hope one may experience in response to seeing an ad about revolutionary dieting pills in one such example, as the active processing of the advertisement stimulus is required to provoke the emotion hope (Poels & Dewitte, 2019). This is in the line with the Theory of Cognitive Appraisal, introduced in chapter 2.2. Extending this theory, the Functional Theory of Emotion states that situational appraisals lead to certain states of action readiness enabling the individual to deal with problems in an adaptive way (Frijda, 1986).

For further examination of emotions in the context of advertisement, a distinction into incidental and integral emotions is proposed (Achar et al., 2016). Integral emotions describe emotions evoked by the content or design of the ad itself and are thereby most of the time deliberately and strategically chosen by the advertising party with a clear manipulative intention. Incidental emotions are evoked by circumstances unrelated to the advertisement itself and are thereby elicited by the context the advertisement is embedded in. Incidental emotions are demonstrated in the findings of Griskevicius et al. (2009), showing that fear induced by a horror movie influences how consumers react to subsequently viewed ads.

In addition, possible interaction effects between incidental and integral effects have also been subject to research. Choi et al. (2016) showed that feelings of loneliness, evoked by prior exposure to images, influence subject responses to negative emotional charity appeals. Berger and Milkman (2012) demonstrated that news articles with predominantly sad content are more likely to result in low activation and less sharing behaviour, whereas for content that predominantly transmitted joy, anger or anxiety, the opposite effect was observable.

Comparison of stimulus constructs

In relation to chapter 2.2, the vital role emotions play in human decision-making is underpinned, as any of the described concepts can be applied to emotions, covering all aspects of the decision-making process.

First, emotions can cover both conscious and unconscious decision processes that are postulated to influence human decision processes by Kahneman. This is demonstrated by the Theory of Cognitive Appraisal and Functional Theory of Emotion for conscious processes on the one hand, and by the Somatic Marker Theory for unconscious processes on the other.

Second, the proposed stimulus response model is observable for emotions, as clear and distinct behavioural patterns are proposed, originating in evolutionary and neuroscientific research.

2.3.4. Construct selection

Resulting from construct evaluation provided by the segment above, emotion is identified as the most promising construct and will be the sole focus of this present work. Table 1 summarizes the decision process. Emotion outperforms Sentiment and Persuasion on the following criteria:

Following the incidental and integral segregation, the potential for both advertisement and its context becomes evident. This is not the case for a construct such as Persuasion, which is solely applicable to the advertisement content. For the given setting, a short mental state like the one evoked by emotions is preferable over a long and enduring one, since the behaviour manipulation needs to take place in a time boxed window, limited by the exposure the advertisement and its context. Furthermore, emotions enable more granular distinctions into different emotion categories, especially compared to the mainly dichotomous classified sentiment. Last, when it comes to predictable reactional patterns, emotion research contains the largest body among the three concepts, providing a strong foundation for environmental stimulus response patterns based on neuroscience and evolution research.

Table 1

Decision criteria for construct selection

Construct	Applicable to	Granularity	Endurance	Response potential
Persuasion	Ad only	6-20 classes	short	high
Sentiment	Ad & context	2-3 classes	enduring	low
Emotion	Ad & context	4-20 classes	short	high

2.4. Emotion classification taxonomies

The identification and choice of an adequate emotion classification taxonomy is important for the final model's performance. The degree of disambiguation and differentiation between the defined emotions the taxonomy provides directly impacts the models' capability to classify the emotions into distinct categories. Plutchik, an early pioneer in psychological emotion research, identified eight basic emotions derived from his Psychoevaluational Theory of Emotions (Plutchik, 1984). The theory claims that emotions served an adaptive role in helping organisms deal with key survival issues posed by the environment. The so-called base, primary or prototypical emotions are characterized by common elements or patterns that are observable in different species. Any other emotion can be derived as mixtures of base emotions (e.g. *love*, *optimism*, *pessimism*). The eight identified base emotions occur in contrastive pairs, underlining the degree of dissimilarity between the emotions, and are often depicted in a circular manner (see figure 1). Plutchik identified the four pairs *joy-sadness*, *fear-anger*, *trust-disgust*, and *anticipation-surprise*.

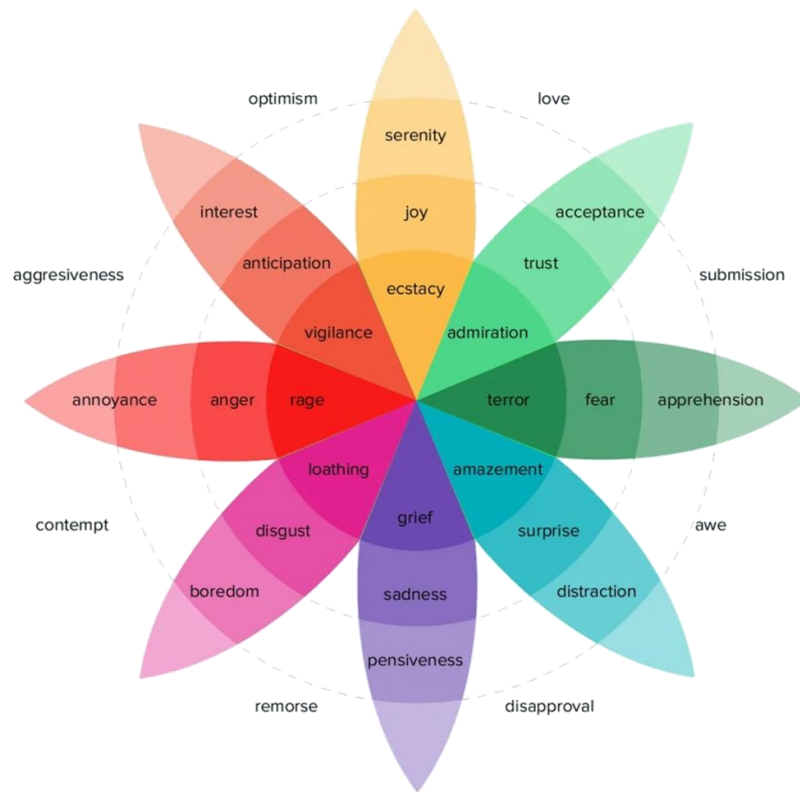


Figure 1. Plutchik's Wheel of Emotion (Sixseconds)

In a similar vein, Ekman (1992) built on the concept of base emotions but departed from the mere theoretical reasoning of evolutionary advantages. Instead, Ekman analyzed and categorized human facial expressions to deduce his set of base emotions, identifying six of them: *anger*, *fear*, *sadness*, *joy*, *disgust* and *surprise*. For practical applications, Liu (2017) findings suggest that for Ekman's six base emotions, disgust empirically yields the lowest predictive performance, which is mostly attributed to the scarcity of samples associated with the emotion amongst the available emotion classification data sets.

For practical and analytical applications, most emotion classification tasks rely on the concept of base emotions and build on variations of either Ekman's or Plutchik's emotion taxonomies. A prominent example is the CrystalFeel (Gupta & Yang, 2018) approach that focuses on emotion extraction based on textual inputs. CrystalFeel limits its consideration to the four emotions *fear*, *anger*, *joy* and *sadness* as they are included in both emotion taxonomies. Furthermore, these emotions encapsule clear adaptive relations to evolution consistent with Plutchik's Psychoevaluational theory of emotions as depicted in figure 2, mapping *fear* to flight and protective behaviour, *anger* to fight and destructive behaviour, *joy* to reproduction and *sadness* to deprivation.

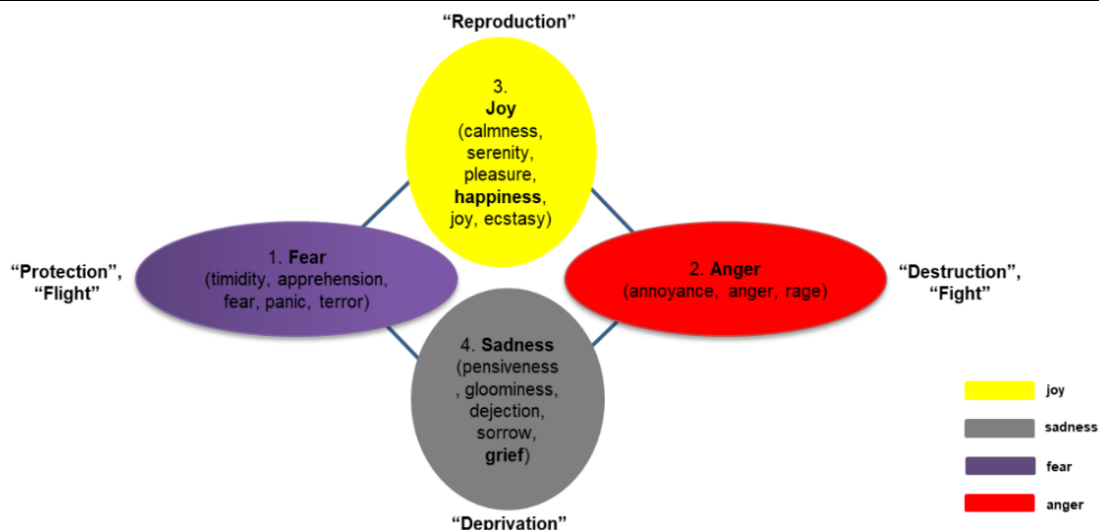


Figure 2. Adaptive relations to evolution for the four base emotions (CrystalFeel, 2018)

Both figures 1 & 2 illustrate that the four base emotions form opposite pairs on Plutchik's wheel of emotion and thereby maximize their differentiation potential, as they are distributed evenly in a 90-degree angle around the wheel. When used for classification, it can be assumed that these emotions yield the best results and are therefore defined and used as the base emotion set in this present work.

2.5. Related work on emotion classification

As described in chapter 1.3., the given data source consists of both texts and images, allowing the usage of both for the classification system's developing. Therefore, the upcoming chapter analyses the current state of the art in emotion classification for text, image and multimodal processing and derives an optimal strategy for the given requirements.

2.5.1. Approaches to text emotion classification

Lexicon-based approaches

Over the past years, a steady refinement of classification methods in the field of NLP is observable. The main challenge is to process the unstructured natural text to identify and extract the best suited features to capture relevant semantics for the classification task. Early work was dominated by lexicon-based approaches (Hartmann et al., 2019) that primarily involve counting the number of words that relate to a certain emotion and determining the overall emotion of a text by deciding on the emotion with the maximum emotion word count. This method relies on emotion lexica that assign one or several emotions to each word present in the lexicon. Thereby, the lexicon-based approach does not necessarily require the training of a machine learning

Related work on emotion classification

model, as the implemented decision rule can be a simple argmax over the distinct emotion classes the lexicon is capturing. Nevertheless, emotion lexica are also useable to derive feature sets that can be fed into supervised learning algorithms as a downstream application.

An example for a popular emotion dictionary is the NRC Emotion lexicon (National Research Council, 2011). The NRC lexicon consists of over 14.000 English words, each assigned to either none, one or a mixture of the eight emotions of Plutchik's emotion taxonomy via crowdsourcing. To date, the NRC lexicon provides the largest released emotion lexicon and was the first of its kind to cover a bandwidth of different emotions in a cohesive way, resulting in its high popularity among NLP research and emotion classification. Consequently, the lexicon has been translated to a range of different languages, including a German version which makes the lexicon applicable for the given context of this work.

Linguistic approaches

In parallel to the emergence of emotion lexica, early work in NLP originated mainly from the field of linguistics. These approaches aim to identify and extract specific linguistic markers that capture useful information for later downstream tasks such as classification. Linguistics markers can be described as corpus statistics, as they accumulate the frequency of specific elements or constructs within a document. Most basic representatives of linguistic features are word frequencies or word positions of specific word groups such as emotion words (see lexicon-based approach), part-of-speech (e.g. verb phrase, noun phrase), complex words (length of a word exceeds a defined threshold), superlatives and so on. Furthermore, grammatical markers are also frequently considered, including punctuation (use of special characters such as question or exclamation marks) and orthography (elongated words in the form of repeated characters, e.g. "I had a reealy tasty meal.", capitalized words, total uppercase letters and so on) (Khlyzova et al., 2022; Gupta and Yang, 2018).

Both linguistics and lexicon-based approaches share the characteristic of being solely based on the frequency of certain markers within a document, meaning that they are basically bag-of-words models that do not take word order, sentence structure or context information into account. For emotion and sentiment detection in text, Kheiri & Karimi (2023) reason that linguistic nuances are crucially important, as the emotion and sentiment transmission in text can occur via subtle textual markers. Examples for linguistic nuances are negations or sarcasm. For negations, the example "*I am not happy today, but I was yesterday*" illustrates the problem a bag-of-word model encounters, as it would identify the emotional marker *happy*, without

Related work on emotion classification

identifying a) the negation that expresses the opposite emotions for today and b) the mixed emotional expression in the covered time span. For sarcasm, a similar problem holds true. For the expression “*Oh great, I just love working on weekends*”, a positive emotional message would be identified by a bag-of-words model, failing to detect the sarcasm that reverses the sentence’s emotional meaning.

These two examples demonstrate the problems of bag-of-words models, as they cannot reliably extract certain linguistic nuances relevant for emotion detection. This is caused by their reliance on word or marker frequencies that fail to integrate contextual information. Therefore, a more complex approach for textual information is required that allows for a context-sensitive representation of unstructured, natural text.

Embedding-based approaches

To handle this issue, text representation in high dimensional, numerical vector spaces is introduced. The core principle of these so-called embedding spaces is to encode the unstructured input data into its numerical vector representation which makes the text interpretable for any machine learning algorithm. In general, sparse and dense embeddings spaces are distinguished. Sparse embeddings are vectors that contain many entries set to zero and only few entries with non-null values. On the other side, dense representations utilize the entire vector length to represent the text (Lee, 2023). This characteristic arises from their differing computation procedures.

Sparse representations work on the actual words present in a corpus by performing some sort of counting or weighting algorithm over all words. Therefore, the vector dimensions remain interpretable, as they capture the occurrence of specific words within the document, which naturally leads to a sparse representation as most words present in the overall corpus do not occur within the encoded document. In the context of NLP, a corpus is the collection of all texts or documents that are available for processing. The most prominent representative of sparse embeddings is the Term Frequency - Inverse Document Frequency (TF-IDF) algorithm. The algorithm is applicable to extracted n-grams from the text corpus.

An n-gram is a sliding window of size n applied to the whole text corpus, extracting all uniquely found windows. Thus, a uni-gram of size one is a resolution to the original word level whereas a bi-gram groups and segregates the processed text into pairs of two neighbouring words. Table 2 illustrates the extraction of different n-grams from a fictional text.

Table 2

Demonstration of fictional n-gram extraction

n-gram	This is a fictional sentence
uni-gram	“This”, “is”, “a”, “fictional”, “sentence”
bi-gram	“This is”, “is a”, “a fictional”, “fictional sentence”
tri-gram	“This is a”, “is a fictional”, “a fictional sentence”

The overall TF-IDF formula is as follows:

$$tfidf_{t,d,D} = tf_{t,d} * idf_{t,D} \quad (1)$$

Where **TF** is:

$$tf_{t,d} = \log(1 + freq(t, d)) \quad (1.1)$$

And **IDF** is:

$$idf_{t,D} = \log\left(\frac{N}{count(d \in D: t \in d)}\right) \quad (1.2)$$

Where t denotes the terms (words), d denotes each document, D denotes the collection of documents.

As formula 1 illustrates, TF-IDF is computed on term level (with terms being equivalents to n-grams) and is the product of the Term Frequency and the Inverse Document Frequency. Term Frequency captures the frequency a term occurs within the current document. Inverse Document Frequency introduces a weighted discount based on the popularity of the term within the whole corpus. In other words, a term that occurs within the current document receives a higher TF-IDF score if it scarcely occurs within the whole corpus, making its appearance in the current corpus more relevant.

The TF-IDF algorithm allows to capture the degree of local word co-occurrence within a document, depending on the chosen n-gram configuration. This can be considered an approximation for the context relevance for the words within the n-gram as related words occur within proximity to each other. Since the TF-IDF approach does not account for word ordering within the n-gram and only captures word frequencies, the embedding lacks the capability to embed word meaning based on its surrounding context.

Related work on emotion classification

To tackle this challenge, more advanced approaches that rely on the training of self-supervised machine learning models on large text corpora have emerged. These models produce dense embedding spaces, capable of representing semantic meaning instead of mere corpus statistics. In principle, the language models are trained on so called “fake tasks” that are only used for the generation of word representations and are discarded after training (Lee, 2023). Two prominent examples are the skip-gram, a task that presents the model with one input word and requires the prediction of its surrounding words, and its inverted counterpart called continuous bag of words, where the model is given all the surrounding words and is required to predict the missing word in the middle.

Early implementations like Facebook’s *fastText* (Bojanowski et al., 2017) relied on shallow neural network architectures. Whilst yielding promising results in embedding word meaning, these models a) still lack the capability to embrace the full context information inherent to surrounding words, especially for longer input sequences and b) are often incapable of disambiguating semantic nuances such as synonyms.

To overcome this hurdle, Vaswani et al. (2017) introduced the Transformer architecture that revolutionized the performance of language models and can be considered the gateway to the current success of NLP applications, as Transformers compose the core of any state-of-the-art language model nowadays. The Transformer architecture, as depicted in figure 3, is composed of an Encoder and a Decoder block. In short, the Encoder extracts features from an input sequence, and the Decoder uses the features to produce an output sequence. Within both Encoder and Decoder, two newly integrated mechanisms are responsible for the observable performance increase for semantic representation extraction.

First, Positional Encoding layers are placed before both Encoder and Decoder blocks, allowing the model to consider the word order of the input sequence. Second, Attention layers form the core of any Transformer block, enabling a simultaneous processing of the whole input sequence. When the sequence is fed into the Attention block, it identifies the sequences’ relevant aspects for the given training task it was trained on, thereby learning context sensitive representations that allow for complex disambiguation. To illustrate the concept Attention, two sentences are given:

- a) “Server, can I have the *check*.”,
- b) “Looks like I just *crashed* the server.”

Related work on emotion classification

When passed to an Attention layer, it would return the amount of attention it puts onto every word in the sentence for solving the training task at hand. For sentence a), the model would learn to give its attention to the word *check*, whereas *crashed* would be the focal point in sentence b). Thereby, the model is capable of disambiguating *server* into a profession in case a) and a machine for case b).

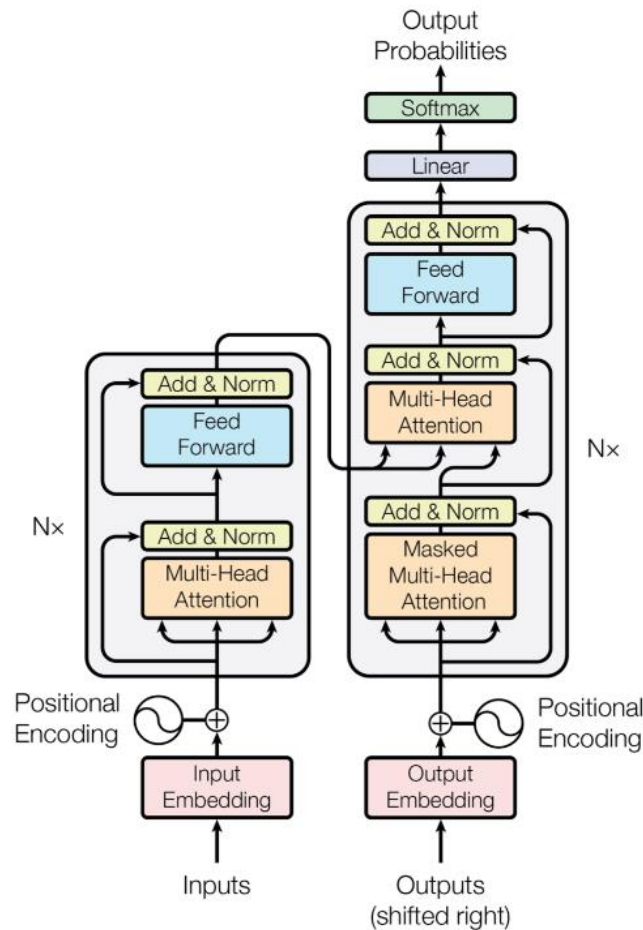


Figure 3. Transformer model architecture (Vaswani et al., 2017)

The application of Transformers in addition to the increase in available computing power and available data led to the emergence of Large Language Models (LLM) in recent years. In this context, large refers to large amounts of both training data and model architectures (that lead to large parameter numbers). One of the first LLM's is the BERT (Bidirectional Encoder Representations from Transformers) model (Devlin et al., 2018) released in 2018 by Google. Trained on a total of 3TB data and including 240 million parameters, the model is bidirectional, meaning it considers both the left and right context. For embedding inference, only the Embedder is used. In parallel, the GPT model architecture was developed by OpenAI (Radford et al., 2018) GPT surpassed BERT in terms of complexity, being trained on 45TB of data in

Related work on emotion classification

consisting of 1.5 billion trainable parameters. Unlike BERT, the GPT architecture consists of a Decoder only. The Decoder concatenates the previous output to the input sequence for every inference, resulting in an autoregressive model architecture. The GPT architecture, more precisely GPT3.5, constitutes the model core for ChatGPT that was launched in 2022 and is considered a milestone in human-like language processing and question answering (Kheiri & Karimi, 2023).

2.5.2. Text emotion classification: Related work

Research interest in emotion classification based on text has been present in the field of NLP for the past decades (Canales & Martínez-Barco, 2014). In accordance with the outlined development in the field of NLP, the integration of embeddings into classification approaches can be noted in recent years. Gupta and Yang (2018) proposed CrystalFeel as introduced in chapter 2.4., that predicts the four base emotions *fear*, *anger*, *sadness* and *joy*. The team used a Support Vector Machine (SVM) and utilized a total of 7100 manually annotated English tweets for training. The proposed approach relies on a feature combination of *fastText* embeddings, TF-IDF uni- and bi-grams, lexicon-based and linguistic features. The resulting model performed best when all feature sets were included, with the lexicon-based feature and *fastText* embeddings yielding the best predictive performance. For the TF-IDF uni- and bigrams, lesser predictive power was reported whereas linguistic features contributed next to no predictive capabilities to the model.

Babanejad et al. (2019) investigated the importance of emotions expressed within 360.000 self-scraped Canadian newspaper articles for predicting user interest in the read article. Therefore, this work does not optimize emotion classification but user interest, which allows for the usage of a large dataset with automatically assigned labels based on captured online interactions with the specific article. Besides TF-IDF embeddings, the model relies on a set of linguistic and lexicon-based features. The conducted experiments indicate that the inclusion of said emotion features contributes substantially to the model's performance in prediction user interest, hinting at a decent performance of the chosen features at capturing and representing emotional content based on newspaper articles.

Delving into the era of transformer-based text processing, Adoma et al. (2020) evaluated BERT's capabilities for emotion classification on the English ISEAR dataset, consisting of over 7000 emotional expressions in a first-person perspective that were generated via crowd sourcing. For the task of classifying seven distinct emotions, BERT outperformed former

classification results on said dataset, demonstrating the promising potential of transformer-based embeddings.

Recently, Kheiri and Karimi (2023) evaluated the performance of GPT embeddings for sentiment classification. Building on the ADA model, the largest and most advanced model OpenAI offers to infer GPT based embeddings, the research team trains Gradient Boosting (GB) and Random Forest (RF) classifiers on 60.000 labelled sentiment tweets. The conducted evaluation revealed that the chosen approach (F1 score: 0.87) outperformed BERT embeddings (F1 score: 0.73) that were utilized as a comparative benchmark substantially. In their future work section, the authors propose further investigation of GPT embeddings, specifically calling for an application to emotion detection.

In conclusion, past work on textual emotion detection has demonstrated the potential of feature combination, especially when using both embeddings and lexicon-derived features (Gupta & Yang, 2018, Babanejad et al., 2019). Amongst the embeddings, transformer-based embeddings show best performance, with GPT embeddings outperforming BERT embeddings for sentiment classification, which allows to formulate the hypothesis of equally superior performance for emotion classification. Lastly, it can be noted that all approaches are based on English datasets with most of them being composed of UGC such as twitter posts. Exceptions like the work of Babanejad et al. (2019) utilize newspaper articles, but do not provide emotion labels.

2.5.3. Approaches to image and multimodal emotion classification

Research on the processing of images and subsequent information extraction is typically referred to as Computer Vision (CV). Early work focused on manual feature extractions, such as identifying regions of interest or colour features like colour averages or histograms (Szeliski, 2022). These features can be fed to a machine learning algorithm for further processing. In similar vein to the advancements in NLP, the emergence of complex neural network architectures had a lasting impact on CV algorithms.

Nowadays, the best performing applications rely on so-called Convolutional Neural Networks, a layer architecture specifically designed to process image data. The CNN design tries to mimic the image processing of the human brain by extracting visual patterns on differing abstracting levels and combining them for object recognition or classification (Szeliski, 2022). Therefore, modern CNN architectures require a sequential stacking of Convolutional Layers, leading to large models that are commonly pretrained on public data and can be fine-tuned for the required task. One prominent example is the Visual Geometric Group (VGG)-Net (Simonyan &

Related work on emotion classification

Zisserman, 2014) that is composed of varying network depth typically between 8 and 16 layers, resulting in up to 140 million trainable parameters. These large CV models output numeric representations of the inserted image and can be used for classification tasks.

For emotion image classification, Yang et al. (2017) retrained a VGG-Net on a the F1 dataset that is a collection of over 23.000 labelled emotion images taken from Instagram and Flickr. For the eight emotion categories present in the F1 dataset, the VGG-Net based showed good capabilities in detecting emotion markers based on images.

Since both image and text data are accessible, exploring possibilities to combine both information sources is advised since “most social media is multimodal” (Yang et al., 2020). Furthermore, a joint consideration of text and image might reveal additional interactional information that a unimodal emotion classifier might miss out on. As an example, Yang et al. (2020) use the image of a crying basketball player. Based on the image only, distinguishing *sadness* and *joy* might be difficult. In this scenario, the image description could provide clarity by reporting on the result of the basketball match which evoked the emotional reaction (tears of *joy* versus *sadness*).

Multimodal information retrieval approaches perform some sort of information fusion to combine both input sources. This is achieved by either applying an early fusion strategy that concatenates text and image embeddings and passes the concatenation to a classifier, or by a late fusion strategy that computes class probabilities separately for image and text and combines both sources for the model’s final prediction (Khlyzova et al., 2022).

Khlyzova et al. (2022) train a multimodal classifier on 1800 text-image pairs scraped from Tumblr posts and manually annotated with Plutchik’s eight emotions. The team combined VGG-Net image and BERT text features to train the emotion classifier. A similar multimodal approach was conducted by Li et al. (2022), also combining BERT embeddings and VGG image features to predict sentiment. Their classifier was trained on the MVSA Multiple dataset, containing over 20.000 image-text pairs scraped from different social media platforms.

2.5.4. Comparison and evaluation

Before implementing the classifier, a decision concerning text and image usage is required. Basing this decision on empirical findings is challenging, as most of the reported work on emotion classification differs in a) data source, b) emotion taxonomy, c) model architecture & applied machine learning algorithm and d) usage of image, text or both. Therefore, reported performance metrics are hardly comparable between the approaches. Consequently, a closer

Related work on emotion classification

consideration of the multimodal approaches is required, as they keep the confounding variables a), b) and c) constant and report results for all three configurations of variable d). Table 3 illustrates reported performance for Khlyzova et al.'s. (2022) and Li et al.'s. (2022) experiments. In both studies, sole image classification performed significantly worse compared to text or multimodal processing. In Khlyzova et al.'s work, text and multimodal processing yielded identical results whereas for Li et al., a slight performance increase is reported for the multimodal setting for sentiment classification.

Table 3

Summary of uni- and multimodal results for emotion classification

Paper	Target	Metric	Image	Text	Multimodal
Khlyzova et al. (2022)	8 emotions	F1 score	0.41	0.53	0.53
Li et al. (2022)	3 sentiments	F1 score	0.61	0.66	0.69

Based on these findings, it is decided to focus on text only for the implementation of the classifier. The minor performance increase for the multimodal is considered too small to justify the integration of multimodal features, especially since the increase is only reported for sentiment analysis, whereas no increase over sole text consideration is found for emotion classification. Besides this empirical argumentation, the required setting described in chapter 1.3. should also be taken into consideration. Since the advertisement is mostly embedded into editorial content in the form of news articles or blog posts, it can be assumed that the reader is primarily set up to process text, derived from his voluntary decision to access the context.

2.6. Emotion classification datasets

To evaluate the potential of emotion text classification, a review of publicly available emotion datasets was carried out. With respect to the given requirements the classifier must meet, the dataset should a) be German, b) consist of PCC and c) relate to either advertisement or publisher content to mimic the prototypical content that is defined by the system requirements. Table 4 summarizes the results of the dataset review. When evaluating the results, two main problems arise:

First, available datasets are almost exclusively published in English, complicating their direct integration into this work. Second, especially larger datasets (*EmoBank*, *CrowdFlower Data*, *GoEmotions*) consist of UGC, mostly in the form of social media posts. These datasets can be labelled automatically, leveraging available metadata such as predefined reactions or hashtags as approximations for emotion categorization (Demszky et al., 2020; Buechel & Hahn, 2017).

For PCC, three news headline datasets were found that are useful to mimic publisher content. The first dataset named *AffectiveText2007* was published in 2007 by Strapparava & Mihalcea and consists of 1050 English news headlines, extracted from major newspapers such as New York Times, CNN and BBC News. The dataset covers six base emotions that were annotated manually by six independent annotators.

The *GerSti* dataset, published by Dang et al. (2021) is the only German dataset found and consists of 1000 news headlines taken from a diverse range of popular German newspapers. *GerSti* includes nine emotions and a neutral class but struggles with severe class imbalance, as the dataset only includes three samples for disgust, nine for shame and 20 for hope, whereas the most popular emotion anger is represented with 226 samples. Emotion labelling was carried out manually through the independent judgement of two annotators.

Lastly, the *GoodNewsEveryone* dataset was published by Oberländer et al. (2020) and contains 5000 English newspaper headlines manually labelled by five annotators into 15 emotion categories. Like *GerSti*, *GoodNewsEveryone* also struggles with class imbalance as negative surprise is the most common emotion with 844 samples, whilst love and pride occur the rarest with only 61 samples for each.

Emotion classification datasets

Table 4

Overview of public text emotion datasets, fitting datasets highlighted in bold.

Name	Author	Taxonomy	Language	Type	Size
ISEAR	Scherer & Wallbott, 1994	7 emotions (joy, fear, anger, sadness, disgust, shame, guilt)	English	UGC: Emotion express.	3000
SemEval 2018	Mohammad et al., 2018	4 emotions (anger, joy, fear, sadness)	English	UGC: Tweets	11.000
EmoBank	Buechel & Hahn, 2017	Valence, Arousal, Dominance	English	UGC: Emotion express.	10.000
GoEmotions	Demszky et al., 2020	27 emotions	English	UGC: Reddit comment	58.000
Affective-Text2007	Strapparava & Mihalcea, 2007	6 emotions (anger, disgust, fear, joy, sadness, surprise)	English	PCC: News headlines	1050
Crowd-Flower Data	CrowdFlower	14 emotions (anger, enthusiasm, fun, happiness, hate, neutral, sadness, surprise, worry, love, boredom, worry, relief, empty)	English	UGC: Tweets	40.000
GerSti	Dang et al., 2021	10 emotions (joy, sadness, fear, disgust, anger, positive surprise, negative surprise, shame, hope, neutral)	German	PCC: News headlines	1000
GoodNews-Everyone	Oberländer et al., 2020	15 emotions (anger, annoyance, disgust, fear, guilt, joy, love, pessimism, negative surprise, optimism, positive surprise, pride, sadness, shame, trust)	English	PCC: News headlines	5000

To the best knowledge of the author, there are currently no emotion datasets publicly available for advertisement texts that would allow to integrate the advertisement content into the classifier. Therefore, it is decided to focus on the context only to deduce the dominant emotion the reader encounters through reading the surrounding article before being exposed to the advertisement. In line with the three identified news headline datasets, the context information will be restricted to its headline.

Besides the data-driven aspect of dataset availability, focusing on the headline for emotion extraction is commonly recommended in emotion detection research. Gupta & Yang (2018) make the argument that the purpose of a newspaper is to catch the reader's interest, which can primarily be achieved by concentrating emotional content onto the headlines. This is in line with the outlined function of emotions to elicit behavioural changes in human behaviour as was described in chapter 2.3.3. This line of thought is supported by Strapparava & Mihalcea (2007) arguing that "headlines typically consist of a few words and are often written by creative people with the intention to "provoke" emotions, and consequently to attract the readers' attention. These characteristics make this type of text particularly suitable for the use in an automatic emotion recognition setting, as the affective/emotional features (if present) are guaranteed to appear in these short sentences".

Furthermore, from a practical perspective, the exclusive consideration of headlines also evades the problem of uncertainty concerning the article's percentage read by the user. Due to the outlined online setting, reading progress cannot be measured and it is well possible that the user skips the part of the article the advertisement is placed in. In addition, since the position of the advertisement is not fixed for all ad slots, it might be disadvantageous to include the whole article unless the advertisement is presented at the bottom of the webpage. For all other cases, it is logical that the user did not read the article in its entirety yet, making the results of a classifier based on the whole article questionable. For headlines on the other side, the probability is very high that the reader read and processed its content, since the reader voluntarily accessed the contents of the webpage. This decision is commonly based on short teasers on referring pages that commonly incorporate the article's headline.

2.7. Synthesis and implications for design & implementation

The design of the implemented classifier is based on the extensive theoretical and empirical evaluation of consumer behaviour prediction in online marketing outlined in the previous sections. First, different options for construct choice were discussed, leading to the choice of emotions as the candidate with the highest potential to predict user behaviour based on its direct linkage to user behaviour that was outlined in chapter 2.3.3 and underpinned with various arguments stemming from social and evolutionary psychology and neurosciences (Ekman, 1992; Plutchik, 1984). Next, it was shown that base emotions are expected to yield the clearest and distinct reactional patterns, specifically the four emotions *fear*, *anger*, *joy* and *sadness* as they form polar opposites on Plutchik's wheel of emotions (Gupta & Yang, 2018).

Delving into related work, it was decided to focus on text only, as the field of NLP provides the largest corpus of related work and results show best performance for text emotion classification together with multimodal classification (see table 3). Ultimately, text was deemed a better fit for the given setup, as especially the publisher content is highly textual by its nature.

The examination of feature strategies indicated that the usage of transformer-based embeddings yields the best results (Kheiri & Karimi 2023) but show the potential of incremental predictive success when combined with sparse embeddings like TF-IDF or lexicon-based features (Babanejad et al., 2019; Gupta & Yang, 2018).

Dataset evaluation showed that the availability of PCC that mimics the required contents is sparse, with only newspaper headlines datasets deemed useful for the purpose of this work. This led to the decision to focus on the context only. Furthermore, as dataset size is small for the three identified datasets *GerSti*, *AffectiveText2007* and *GoodNewsEveryone* and two of them are in English, the ensuing implementation requires a strategy to deal with the language barrier and dataset size.

3. Design and Implementation

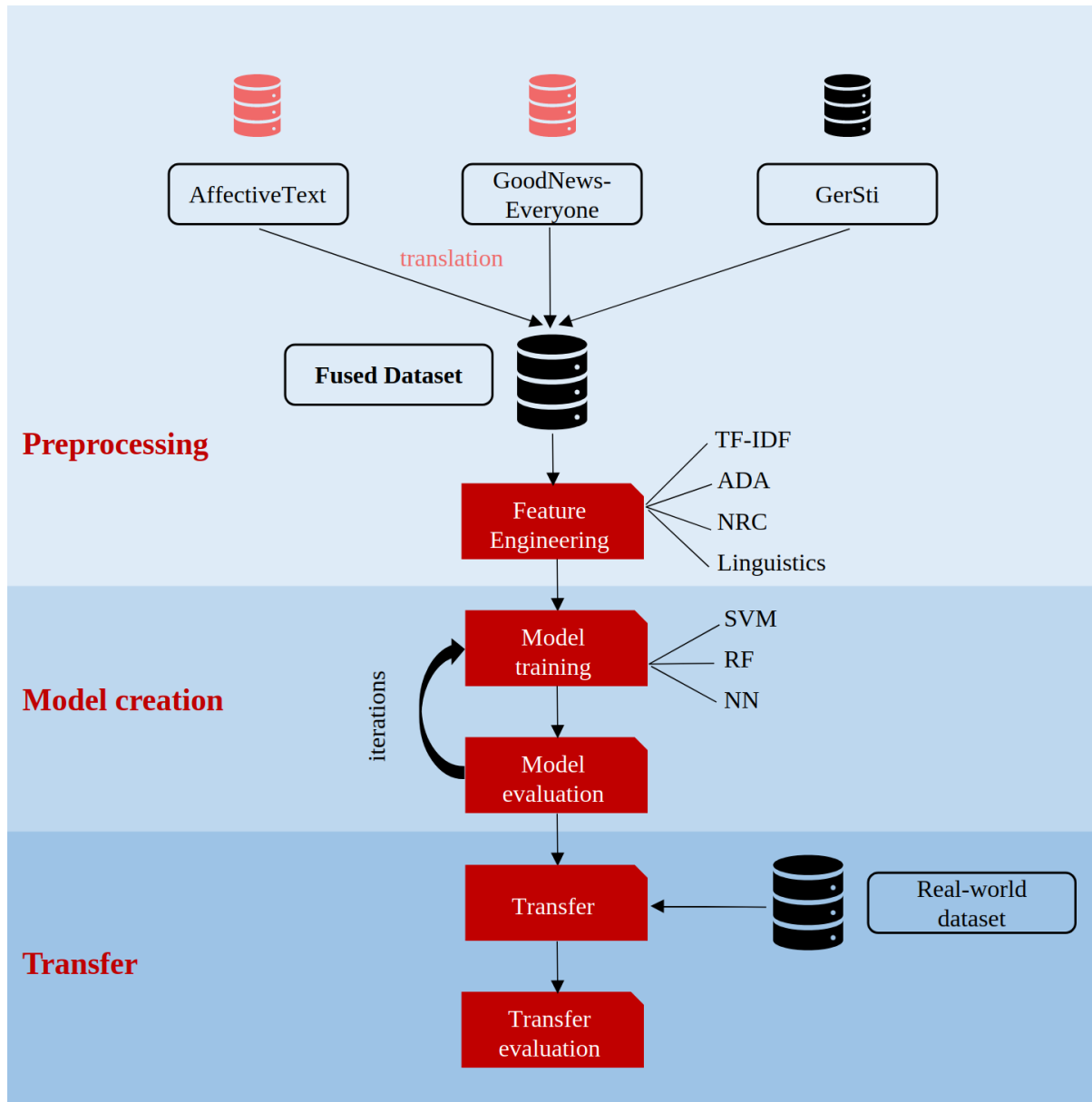


Figure 4. Overview of implemented learning framework

Figure 4 illustrates the learning framework that was built to train, evaluate and deploy the emotion classification models. The first section *Preprocessing* handles the cleaning and combination of the three data sources to provide a fused dataset. Next, feature engineering is performed to extract the required features from the raw input text. Lastly, data is split into train, validation and test data with two sequential splits using an 80:20 ratio, resulting in the final data split shown in figure 5.

Preprocessing (Section 1)

Section 2 iteratively tests different model architectures, varying feature combinations and learning algorithms to select a final model configuration for which final performance metrics on the test set are reported.



Figure 5. Used data split into train, validation and test sets

Section 3 transfers the trained classifier to the main corpus, a collection of real-world contextual contents that are enriched with information on user’s interaction rates, allowing to evaluate the impact of detected emotionality in the context on user behaviour.

The framework is written in python, with model training executed using the scikit-learn and TensorFlow libraries.

3.1. Preprocessing (Section 1)

3.1.1. Dataset creation

As outlined in section 2.6., the three datasets *AffectiveText2007*, *GoodNewsEveryone* and *GerSti* are taken into consideration as they all classify emotions on news headlines. Since *GerSti* is the only dataset available in German, a translation strategy for the other two datasets is required. For this task, an evaluation strategy for the translation quality is needed, since it is of crucial importance that semantic nuances are not lost during the translation process and to guarantee a maximization in semantic representation.

For scientific application, it is common practice to employ a retranslation strategy (Sahin & Dungan, 2014). After the translation into the target language, a retranslation into the original language is performed. Every pair of original retranslated text is consequently evaluated manually by an expert on semantic coverage and similarity. For the present work, relying on manual evaluation is unfeasible for two reasons: First, the expert’s judgement is subjective, as it does not follow objective decision rules and is mostly based on the expert’s language understanding. Second, for large datasets manual evaluations are a cumbersome endeavour that requires substantial resources to execute.

Therefore, an automatic evaluation of translation results is proposed that applies the principles of the retranslation approach. In a first step, translations from the English original to German are done using a python API to access the Bing translation service. The resulting German

Preprocessing (Section 1)

headlines are retranslated to English and for both original and retranslated text, BERT embeddings are computed. As outlined in chapter 2.5.1., as a transformer-based model BERT is a well-suited model to capture contextual semantic meaning of text and is therefore considered a good fit for the required task. To replace the expert's judgement, embedding similarities for every pair of original and retranslated embedding vectors are computed and tested against a threshold.

This approach is based on the idea that semantic similarity in an embedding space is represented by vector distance and orientation. For two identical sentences, their respective vectors are equal, resulting in a maximized embedding similarity. Therefore, this approach allows to capture semantic representations through the usage of BERT embeddings and quantifies semantic similarity as embedding space distance. To measure embedding similarity, cosine distance or similarity is utilized, as it maps the angle between the compared vectors to a range of 0-1, with 0 representing diametral vector orientation (180°) and 1 representing identical vector orientation (0°). Cosine similarity is computed using the dot product and the absolute values of the vectors:

$$\text{cosine similarity} := \cos(\theta) = \frac{A \cdot B}{||A|| ||B||} \quad (2)$$

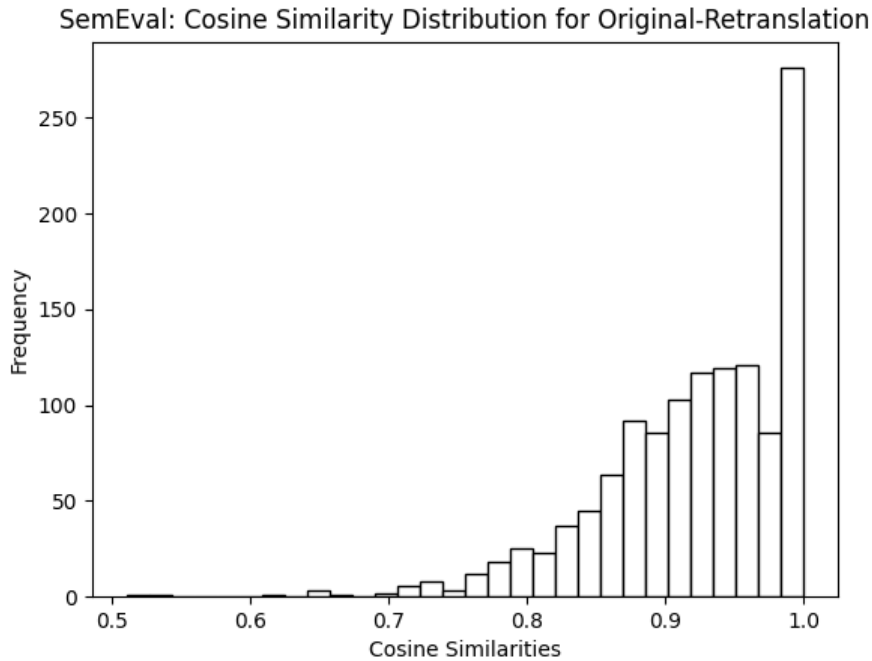


Figure 6. Cosine similarity distribution for original-retranslation text pairs on the *AffectiveText2007* dataset

Preprocessing (Section 1)

Figures 6 and 7 show the resulting cosine similarities for both datasets. The heavy skewedness for both distributions indicates good translation results, with most headlines achieving cosine similarities above 0.85. To improve dataset quality, any sample with a cosine similarity below this threshold of 0.85 were excluded from the resulting dataset, which led to a selection of 86,4% of all cases for the *AffectiveText2007* dataset and 94% of cases for the *GoodNewsEveryone* dataset. For both datasets, the case removal increased mean cosine similarity slightly from 0.92 to 0.94 for *AffectiveText2007*, and from 0.939 to 0.947 for *GoodNewsEveryone*.

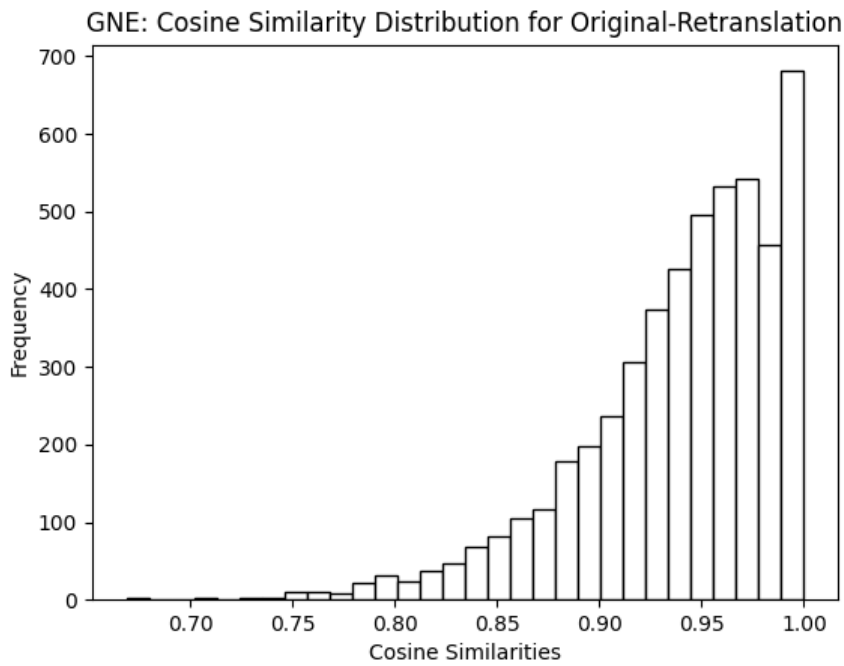


Figure 7. Cosine Similarity Distribution for original-retranslation text pairs on the *GoodNewsEveryone* dataset

To merge the resulting German data source, the different emotion sets are combined to create a holistic dataset. From the resulting merge, the emotions *shame*, *pride* and *trust* are discarded, as they barely occur in the merged dataset. The resulting dataset is described in table 5. The neutral *no emotion* category is explicitly included, providing more learning flexibility to the algorithm as this class prevents the classifier from assigning an emotion to every news headline. Regarding its later application to real-world data, this is considered important to account for an expectably larger number of neutral headlines. This assumption is based on the manual selection of headlines in the training data to approximate class balance, which is not expected to be found in real-world application but an important requirement for machine learning applications.

Preprocessing (Section 1)

For the upcoming evaluation and testing, the complete set as well as a reduction to the four base emotions is utilized, to provide benchmark for the evaluation of the base emotion's effectiveness.

Table 5

Resulting class distribution for merged dataset, base emotions highlighted

Emotion	Absolute	Relative
No emotion	501	0.079
Anger	707	0.111
Fear	714	0.112
Joy	693	0.109
Sadness	613	0.096
Positive Surprise	608	0.096
Negative Surprise	986	0.115
Annoyance	397	0.062
Disgust	375	0.059
Guilt	162	0.025
Pessimism	300	0.047
Optimism	297	0.047
Total	6353	1

3.1.2. Feature Engineering and preprocessing

In a first step, the raw input text is pre-processed by separating input texts to word level (also called tokenization). On token level, stop words and special characters that do not carry semantic meaning such as parenthesis and quotation marks are removed. Lastly, words are reduced to their lemma, which is the process of reducing a word to its dictionary root form, and are converted to lowercase. This procedure is required for operations that rely on count operations. Without lemmatization, the words *run* and *running* would not be recognized as equal, whereas their lemma *run* aggregates both cases.

TF-IDF. To provide a baseline of embedding classification performance, TF-IDF features are extracted by computing uni- and bigrams from the training corpus. To limit the resulting feature space, the 100 most frequent uni- and bigrams are selected.

ADA. Following the advice of Kheiri & Karimi (2023), ADA embeddings are inferred using the OpenAI API to access the *text-embedding-ada-002 model* which returns a 1536-dimensional embedding representation of text input. Importantly, the ADA features are extracted on the raw text before the described preprocessing is executed. This is the recommended procedure, as the ADA model leverages contextual clues through its transformer-based architecture. Thus, pre-processing might negatively affect the model's performance. More importantly, the model was trained on raw text data, meaning that best embedding representations are to be expected for unprocessed inputs.

NRC. As Babanejad et al. (2019) and Gupta & Yang (2018) demonstrated, the integration of lexicon-based and linguistic features can improve classifications results over sole embeddings. Therefore, the German version of the NRC lexicon mentioned in chapter 2.5.1. is used to extract an NRC feature set. The general procedure for lexicon-based feature extraction is to consider every word of an input sequence separately by looking for word matches within one of the lexicon's categories. Lastly, an aggregation function like a summation or an average is applied to the whole sequence of words.

To increase information gain from the NRC lexicon, the basic approach of counting words present in the lexicon is extended by an embedding based approach. Therefore, *fastText* embeddings are computed for every word of the NRC lexicon as well as for every word of the input sequence. *FastText* embeddings, as shown in chapter 2.5.1., allow for good semantic representation on word level and are consequently chosen. The extracted *fastText* embeddings of NRC words are averaged per class of the NRC lexicon, providing an average representation

Preprocessing (Section 1)

of said class within the embedding space (e.g. a representative vector for the emotion *fear*, computed as the average over all words the lexicon assigns to the emotion). For every word of the input sequence, the cosine similarity to the NRC class average embedding is computed and used as a feature. Compared to the sole word count approach, leveraging embedding features captures semantically related words and represents them based on their proximity to the class centroid, even if the words themselves are not part of the lexicon.

The NRC lexicon is composed of ten classes and assigns class values to every word accordingly. The classes are the eight emotions of Plutchik’s taxonomy as well as two classes that capture positive or negative word connotation. The lexicon also captures emotional ambiguity as it allows for multiple classifications per word. As an example, the word *verlassen* is assigned to the classes *negative*, *sadness*, *fear* and *anger*.

To extract features, relative frequencies of emotion words for each of the ten categories are counted, accounting for different headline lengths and normalizing the resulting feature to a solution space of 0-1. Next, the cosine similarity for every input word and the *fastText* class centroid is computed and averaged over the input sequence. Thus, 20 features are extracted.

To capture the information on mixed emotionality, four new classes are computed. The first class evaluates whether a word is purely assigned to one emotion by the NRC, whereas class two measures if any emotion assignment is present. Class three and four evaluate if a word is assigned to at least two or at least four emotion classes simultaneously. For the four new classes, the same feature extraction approaches based on class count and embedding centroid are applied, increasing the feature size to 28.

Lastly, the position of the first occurrence of a positively or a negatively labelled word is measured, resulting in a final feature size of 30.

Linguistics. As relevant linguistic markers for emotion detection, the count of exclamation and question marks are evaluated separately. Additionally, sentence length defined as total word count, as well as the average word length per headline is included to the feature set, resulting in a total of four linguistic features.

The summary of the ultimate feature sets is given in table 6.

Table 6

Summary of feature sets

Feature Set	TF-IDF	ADA	NRC	Linguistics	Total
Features	1000	1536	30	4	2570

3.2. Model creation (Section 2)

To find an optimal classifier, feature and model combinations are tested iteratively, with decisions on further model refinement being made after each iteration.

3.2.1. Training iterations

1st iteration

Based on results of related work, random forest (RF) and support vector machine (SVM) models are implemented as they traditionally perform the best for text emotion classification (Gupta & Yang, 2018; Gambino & Calvo, 2019; Kheiri & Karimi, 2023).

An SVM is a supervised machine learning algorithm that classifies data points by finding the hyperplane that maximizes the margin between classes. This hyperplane is defined by the of the so-called support vectors, given the algorithm its name.

A Random Forest is an ensemble learning method for classification that constructs multiple decision trees during training and outputs the mode of the classes as its prediction.

For this first iteration, the two classifier models are tested for all four described feature sets. Experiments are conducted for both base and complete emotion sets, resulting in a total of 16 experiments.

2nd iteration

The second iteration focuses on comparing feature set combinations to identify potential incremental effects when augmenting embeddings with other feature sets, as postulated by the works of Babanejad et al. (2019) and Gupta & Yang (2018). Restricting itself to only one classifier, the second iteration consists of 5 experimental trials per emotion set.

3rd iteration

The third iteration intends to handle observed overfitting during the first two iterations. Overfitting in the context of machine learning describes a model's tendency to perform

Model creation (Section 2)

significantly better on training data than on the unseen validation data. Overfitting is typically caused by three main sources (Géron, 2023):

Model complexity. A first source of overfitting occurs when a model fails to learn overarching patterns and characteristics to differentiate classes, but rather mimics the exact shape of the training data. This problem is comparable to a student that learns a subject's matter by heart and performs exceptionally well at an upcoming test but fails to transfer general concepts of the matter to new applications. To prevent this type of overfitting, example approaches are to remove features from consideration or to introduce regularization terms that punish the model for putting too much attention onto specific features.

Data Scarcity. Second, the available data source may not be sufficient to cover all nuances that are relevant to distinguish classes properly. In this case, increasing dataset size is mandatory to increase classification performance.

Data Quality. Lastly, bad data quality also directly influences overfitting. Examples are noisy data that include many erroneous examples and outliers, or inaccurate class labels. Thus, a thorough revision of dataset quality issues can contribute to the mitigation of overfitting.

As overfitting sources two and three are out of control during model training, iteration three intends to tackle model complexity by hyperparameter tuning the best model found in iteration two as a first step, and secondly by implementing a shallow neural network (NN) architecture. For the NN, hyperparameter tuning is performed to explore the versatile options the NN architecture provides for overfitting prevention. Examples are the inclusion of dropout layers, early stopping implementations, varying layer sizes and numbers, batch normalization and the usage of varying learning rates (Mitchell, 1997).

The best performing model found after the third iteration is evaluated on the test dataset to report final performance metrics for both complete and base emotion sets.

3.2.2. Evaluation strategy

Score selection. When it comes to the selection of an evaluation metric, most work on multiclass emotion classification report the accuracy and the F1 score. While accuracy considers the ratio of correct classifications on all classifications, the F1 score combines and balances the precision and recall influence on the overall classification performance as shown in formula 3:

$$F1 = 2 * \frac{Precision * Recall}{(Precision + Recall)} \quad (3)$$

Model creation (Section 2)

Precision is computed as the ratio of true positives amongst all positively classified instances (true positives + false positive), whereas Recall is the ratio of true positives on all actual true instances (true positives + false negatives). For the given setting, the F1 score is chosen as the ultimate decision metric, as it handles class imbalance, which is not given for the accuracy measure.

To compute an F1 score (and thereby Precision and Recall) in a multiclass setting, scores must be computed on class level and averaged subsequently. To average the F1 score, either a micro or a macro strategy may be deployed. The micro strategy computes an overall confusion matrix (true positives, false positives, true negatives, false negatives) over all classes and computes recall and precision scores based on the overall matrix. The macro strategy on the other hand computes Recall and Precision scores separately per class and takes an average over the class scores. For the experienced class imbalance, a micro strategy is used as it accounts for class imbalance, whereas the macro strategy does not as it gives an equal weight to each class at final score computation (Géron, 2023).

Baseline comparison. To evaluate performed experiments, results provided by related work is usually taken for comparison. For this work, comparability with former work is hardly given, due to the unique creation of a merged dataset, that on the one side includes a neutral class which is seldomly the case in other approaches and on the other side is in German, for which baseline results are scarce anyways. In general, comparability for emotion classification is problematic, as there is no unified taxonomy of emotion classes, leading to different emotions classified on different datasets. The TF-IDF results are considered as the baseline it forms the basis for a variety of past work on emotion classification (Babanejad et al., 2019; Gupta & Yang, 2018).

3.3.Transfer (Section 3)

In a final step, the resulting emotion classifier is applied to unseen real-word data to evaluate its potential for consumer behaviour prediction.

Transfer of the classifier. First, a dataset of approximately 4500 headlines taken from context articles that were released on German publisher websites are passed to the trained classifier. These articles are exemplary articles that were used for advertisement placements. As this dataset, subsequently referred to as main corpus, is unlabelled by nature a quantitative evaluation is not possible. Instead, distributional descriptives are provided and exemplary headlines are shown and analyzed to evaluate the classifier's capacity on unseen and real-world headlines.

Consumer behaviour prediction. For every classified context in the main corpus, all measured user interactions are allocated. These session instances are only measured if a so-called impression occurs. For a session to count as an impression, the user must be given the chance to become aware of the advertisement inherent to the context. Thus, if a user opens a context article without scrolling to be part the advertisement is placed in, the session is not considered an impression. For every measured impression instance, user interaction with the advertisement is measured by a click onto the advertisement canvas. The average click rate over all session is called the Click-Through-Rate (CTR) and represents the percentage of all instances that lead to a click onto the advertisement.

To evaluate the impact of emotions in contextual design, the main corpus is grouped by the emotion categories and therefore by the measured impressions per emotion category. Based on this data aggregation, CTR differences between the emotion classes can be analyzed empirically.

4. Results

4.1. 1st iteration: Baseline evaluation

Following the outlined experimental setup described in the above chapter, the first set of experiments evaluates the potential of different machine learning algorithms in combination with different feature sets applied in isolation. Tables 7, 8 and figures 8, 9 summarize the experimental results for both basic and complete emotion sets.

Table 7

Baseline results for base emotions for different algorithm X feature combinations

5 classes		Train		Validation	
Feature	Model	F1	Accuracy	F1	Accuracy
Linguistics	SVM	0.246	0.259	0.248	0.258
	RF	0.557	0.558	0.220	0.221
NRC	SVM	0.610	0.617	0.412	0.426
	RF	0.999	0.999	0.433	0.438
TF-IDF	SVM	0.923	0.922	0.430	0.438
	RF	0.952	0.952	0.434	0.442
ADA	SVM	0.981	0.981	0.587	0.615
	RF	0.999	0.999	0.558	0.578

The conducted experiments differentiate the feature's potential for emotion detection clearly and for both RF and SVM. The ADA embeddings outperform TF-IDF embeddings and NRC-based features significantly, with the two latter sets performing equally. The isolated usage of linguistic features did not provide any predictive potential as they barely outperform a random guess and are consequently discarded from further consideration. Both TF-IDF and ADA show good capabilities to capture emotional nuances, which is represented by their high predictive

1st iteration: Baseline evaluation

performances on the train set. NRC and linguistic features appear to be too simplistic to represent all emotional nuances, indicated by lower results on the training data.

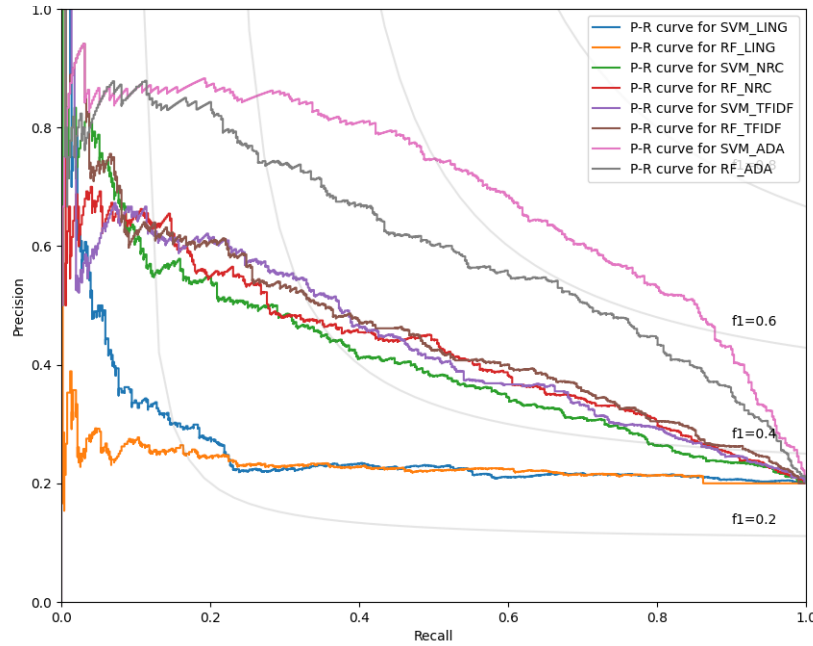


Figure 8. Base set Precision Recall curves for baseline experiments

Comparing SVM and RF performance, the SVM implementation outperforms RF for ADA embeddings (Validation F1: 0.59 vs 0.56 for base and 0.36 vs 0.28 for complete set). On the other hand, for NRC features and TF-IDF embeddings, the RF implementation slightly outperforms SVM (see tables 7 & 8). These findings are supported by the provided Precision Recall curves (see figures 8 & 9). The curves evaluate the trade-off between Precision and Recall for varying thresholds, allowing for a fine-grained evaluation of the provided model's performance. The performance difference between SVM and RF for ADA embeddings is represented by the gap between the grey line for the RF and the pink line for the SVM implementation for both complete and base emotion sets. Also, the smaller differences for TF-IDF and NRC between the two models are represented by neglectable gaps between brown and purple as well as between green and red lines in both figures.

1st iteration: Baseline evaluation

Table 8

Baseline results for complete emotions for different algorithm X feature combinations

12 classes		Train		Validation	
Feature	Model	F1	Accuracy	F1	Accuracy
Linguistics	SVM	0.110	0.117	0.110	0.117
	RF	0.403	0.407	0.132	0.136
NRC	SVM	0.672	0.673	0.231	0.241
	RF	0.999	0.999	0.232	0.254
TF-IDF	SVM	0.906	0.903	0.227	0.240
	RF	0.936	0.933	0.236	0.247
ADA	SVM	0.996	0.996	0.357	0.391
	RF	0.999	0.999	0.280	0.324

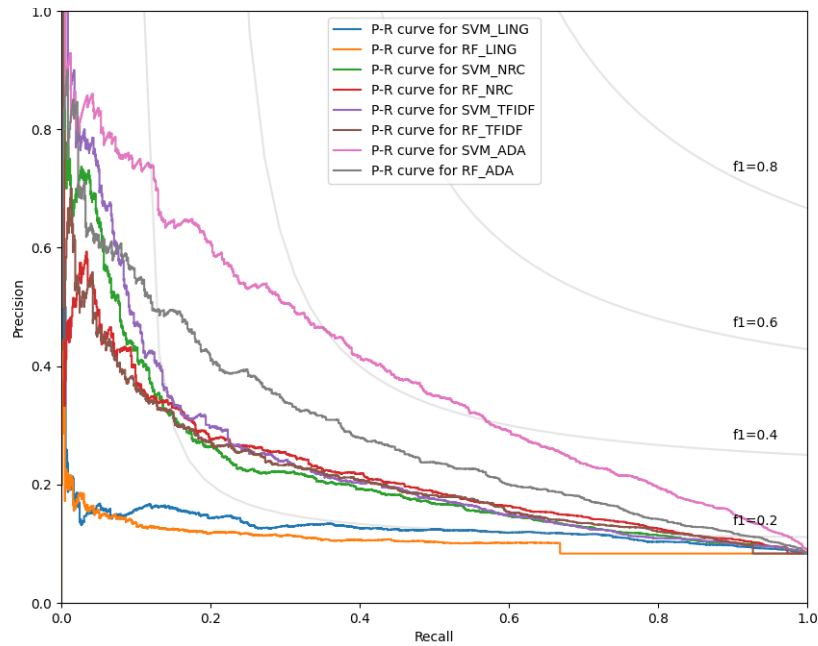


Figure 9. Complete set Precision Recall curves for baseline experiments

2nd iteration: Feature combinations

Summarizing these findings of all baseline's experiments, three major aspects are noted:

Firstly, SVM's are considered superior for the given emotion classification task, especially based on their significant outperformance when utilizing ADA embeddings which yielded the best classification results.

Secondly, high overfitting in the form of a large gap between train and validation results is noted over all baseline experiments (see tables 7, 8).

Thirdly, the basic set, comprised of five classes, performs significantly better compared to the complete set that includes twelve categories (Average F1 of 0.42 for basic setting versus 0.23 for complete setting). These findings are partly to be expected due to the larger number of classes, but call for further investigation by examining and contrasting individual class performance to gain further insights, which will consequently be carried out after a final model is selected and trained in chapter 4.4.

4.2. 2nd iteration: Feature combinations

Building on the baseline evaluation's findings, the second iteration tests the potential of incremental classification results by combining the presented feature sets. These combinational trials are trained using an SVM, showing the best overall performance in the first iteration.

Table 9

Feature combinations versus ADA baseline for base emotions

5 classes		Train		Validation	
Feature	Model	F1	Accuracy	F1	Accuracy
ADA		0.981	0.981	0.587	0.615
TF-IDF & NRC		0.935	0.935	0.451	0.461
ADA & TF-IDF	SVM	0.985	0.985	0.600	0.627
ADA & NRC		0.980	0.980	0.601	0.625
ADA & TF-IDF & NRC		0.985	0.985	0.602	0.627

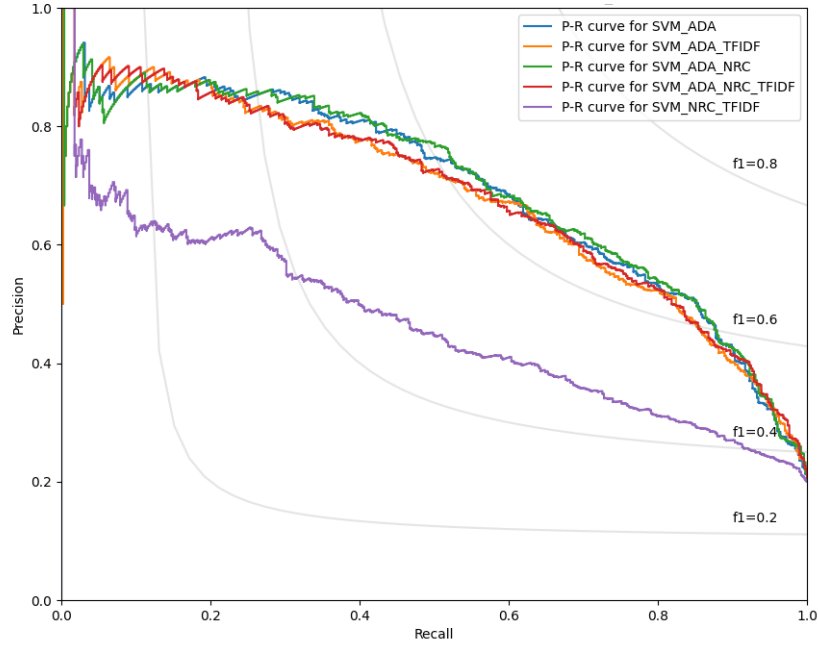


Figure 10. Base set Precision Recall curves for feature combinations

Tables 9 & 10 summarize trial results for base and complete emotion settings and contrast them to the best performing SVM model trained on ADA features solely found in the 1st training iteration. For the base setting, there is a notable increment for ADA feature expansion, with all three expansion sets achieving approximately an F1 of 0.6 compared to the 0.587 of the mere ADA feature utilization. The combination of TF-IDF and NRC features clearly lack behind any ADA feature combination, but still show significant increases over sole TF-IDF and NRC applications (compare table 9). These findings are visualized in figure 10, with purple representing the TF-IDF and NRC combination. For the ADA combinations, the Precision Recall curve demonstrates a differentiation between the green (SVM & ADA) and blue (ADA) lines and the red (ADA & NRC & TF-IDF) and orange (ADA & TF-IDF) lines with the later pair showing slightly worse classification performance. For the upper group, the SVM and NRC combination yields slightly better results.

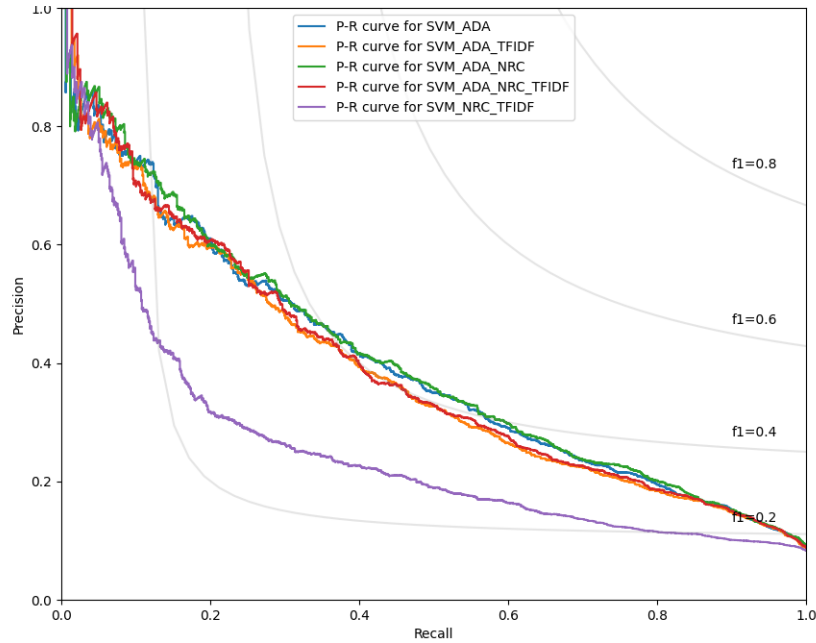
For the complete setting, the above-described patterns are replicable as shown in figure 11 and table 10, with and ADA and NRC combination performing best.

2nd iteration: Feature combinations

Table 10

Feature combinations versus ADA baseline for complete emotions

12 classes		Train		Validation	
Feature	Model	F1	Accuracy	F1	Accuracy
ADA		0.996	0.996	0.357	0.391
TF-IDF & NRC		0.929	0.928	0.254	0.270
ADA & TF-IDF	SVM	0.997	0.997	0.343	0.380
ADA & NRC		0.996	0.996	0.357	0.391
ADA & TF-IDF & NRC		0.997	0.997	0.350	0.385

*Figure 11. Complete set Precision Recall curves for feature combinations*

Derived from the 2nd iteration, the combination of ADA and NRC features is chosen as the most promising feature set for both complete and base emotion sets. Just like for the 1st iteration, notable overfitting is detected in the second trial. Therefore, a 3rd and last iteration will explicitly handle the issue of overfitting.

4.3. 3rd iteration: Finetuning and overfitting

To handle overfitting across all models of the 1st and 2nd iteration, two approaches are tested on the chosen ADA and NRC feature set: Firstly, hyperparameter tuning is carried out for the SVM model. Secondly, a Neural Network is trained due to its flexibility in handling overfitting as outlined in chapter 3.2.1.

SVM hyperparameter tuning. To identify optimization potential for the SVM application, a grid search was performed using the grid included in Appendix A. For both base and complete emotions sets, the best performance was achieved by only adapting the C parameter, changing its value from 1 to 20. This resulted in minor improvements for the base setting (F1 increment of 0.601 to 0.602) but yielded equal results for the complete set.

Neural Network testing. To explore the potential of neural networks, hyperparameter search was carried out for both emotion settings. Since model complexity does not seem to account for overfitting as demonstrated by the SVM tuning, a shallow architecture is chosen and tuned. Consequently, variations of two and three fully connected hidden layers are tested with varying layer sizes, learning rates and batch sizes (for full grid, see Appendix B).

Figures 12-15 illustrate hyperparameter tuning results for both settings, plotting the development of loss and F1 score over trained epochs for both train and validation data.



Figure 12. Loss over training epochs for train and validation set on base emotions

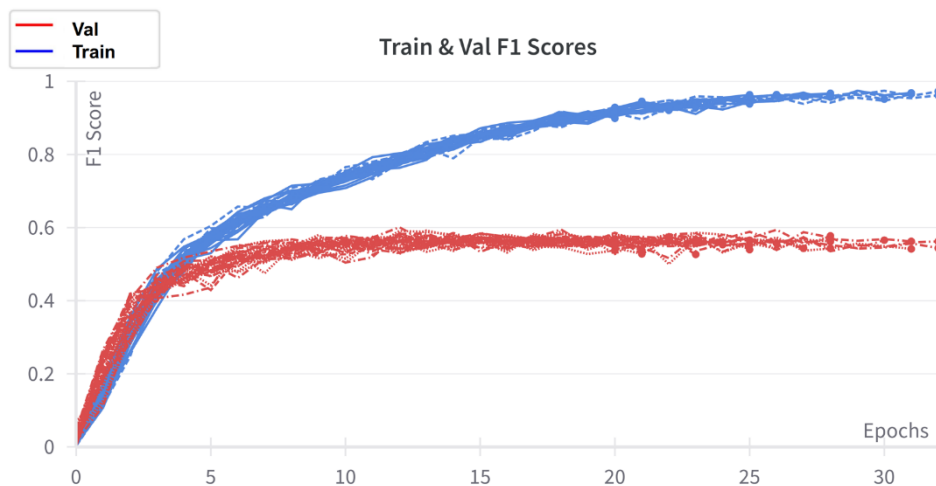


Figure 13. F1 Score over training epochs for train and validation set on base emotions



Figure 14. Loss over training epochs for train and validation set on complete emotions

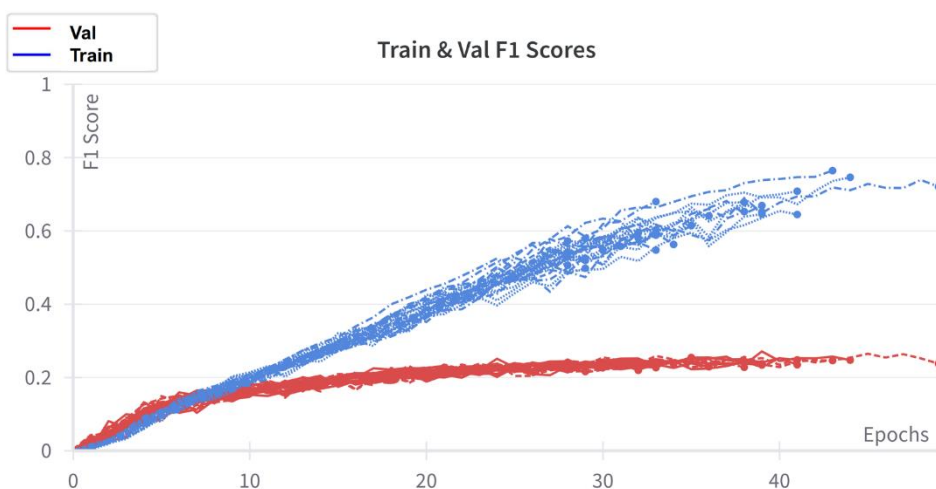


Figure 15. F1 Score over training epochs for train and validation set on complete emotions

3rd iteration: Finetuning and overfitting

In both cases, the same tendency of initially decreasing training and validation loss can be observed (figures 12 & 14). After a few epochs, the validation loss starts to increase again whilst training loss keeps decreasing until convergence which is the expected pattern for overfitting. Notably, the divergence of train and validation loss starts at a later epoch for the complete setting (around 15th epoch) compared to the base setting (around 8th epoch).

The loss evaluation is consistent with examinations of the F1 development over the epochs (figures 13 & 15) with steady increases in training F1 performance, whilst validation F1 scores plateau once train and validation loss divergence starts at the mentioned epochs for each emotion setting respectively.

In summary, the performed hyperparameter tuning resulted in very slim performance differentiation, illustrated by the narrow “tubes” that the joint consideration of related metric threads over different model configurations form. To deduce the best model, the development of the F1 score over the training epochs was monitored and the best result on the validation set was extracted. The resulting model architectures for both emotion settings are summarized in Appendix C.

Table 11 compares the results of the 3rd training iteration, showing better performance for the NN model. This finding is supported by figure 16, illustrating that the NN model outperforms the tuned SVM for the base setting (blue and green lines) as well as for the complete setting (orange and red lines).

Table 11

Final comparison of tuned models

Feature	Model	Classes	Train		Validation	
			F1	Accuracy	F1	Accuracy
ADA & NRC	Tuned SVM	base	0.980	0.980	0.602	0.626
	NN		0.820	0.820	0.617	0.621
	Tuned SVM	complete	0.996	0.996	0.357	0.391
	NN		0.998	0.998	0.357	0.361

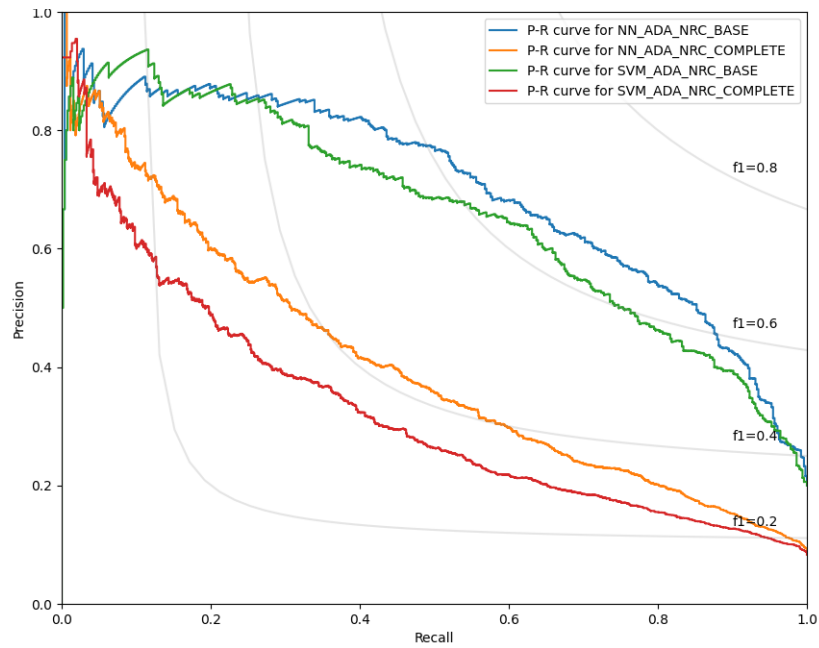


Figure 16. Precision Recall curves for tuned SVM and NN for both emotion sets

Consequently, a NN model on ADA and NRC features are selected as final models for both emotion sets.

4.4. Evaluation on test set

As a last evaluation step, the resulting model configurations are retrained on the test set and compared to former baselines to provide final performance metrics. Figure 17 illustrates the superior performance of the final NN implementation compared to a TF-IDF baseline approach, like the one Babanejad et al. (2019) proposed, and to the best performing SVM as an extended baseline result.

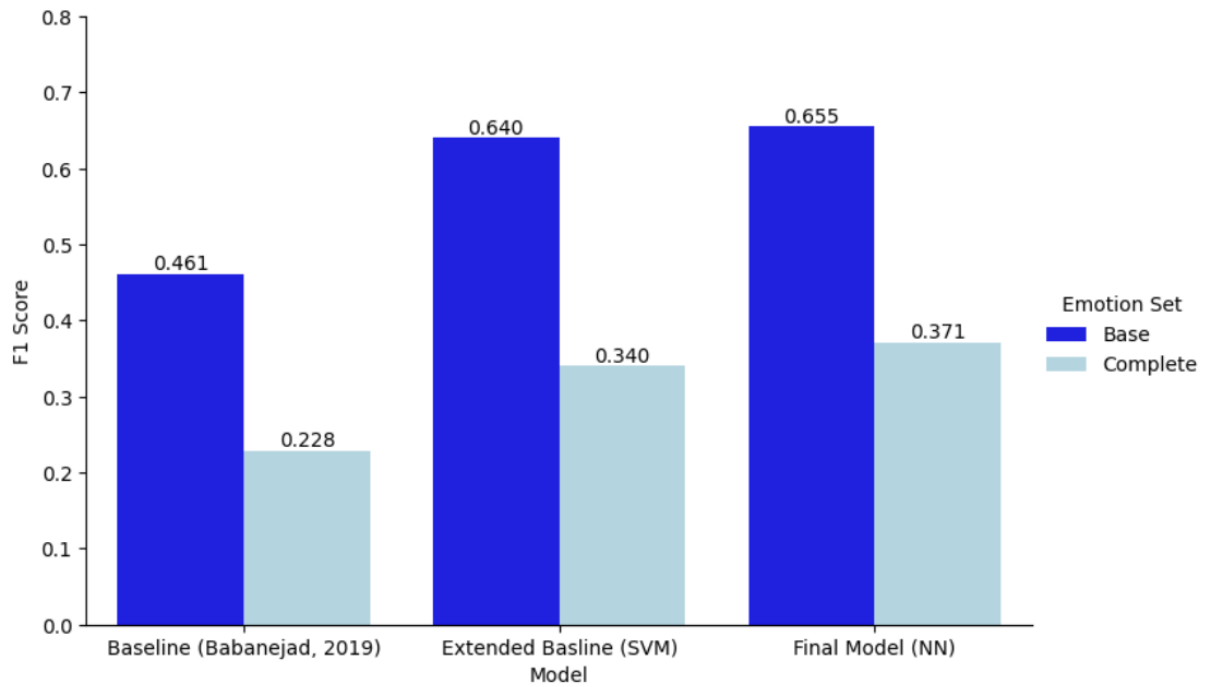


Figure 17. Final performance comparison on test data

Detailed insights on model performance of the final model are provided in table 12. For the base setting, the test set results surpass achieved validation scores (total F1 of 0.655 on test set, 0.617 on validation set), whereas a slow performance decrease is observable for the complete setting (F1 of 0.35 on test set, 0.357 on validation set).

For a detailed analysis of the model's capability to classify emotions, figures 18-21 compare classification performance between emotion classes.

Table 12

Classification performances for final model on emotion level

	Base emotions		Complete emotions	
	Train F1	Test F1	Train F1	Test F1
No emotion	0.926	0.511	0.996	0.378
Anger	0.810	0.667	0.999	0.441
Fear	0.789	0.685	1	0.485
Joy	0.862	0.758	0.999	0.479
Sadness	0.768	0.618	0.993	0.433
Positive Surprise	-	-	0.999	0.275
Negative Surprise	-	-	0.993	0.288
Annoyance	-	-	0.988	0.131
Disgust	-	-	0.988	0.203
Guilt	-	-	1	0.310
Pessimism	-	-	0.998	0.155
Optimism	-	-	0.998	0.210
Total	0.786	0.655	0.996	0.350

For base emotions, table 12 and figure 19 indicate that the neutral category appears to be the hardest to distinguish, resulting the lowest F1 score on the test set. The three emotions *sadness*, *fear* and *anger* perform equally well with a F1 score in a range of 0.62- 0.67. The emotion *joy* performs the best with and F1 score of 0.76. These findings might be evoked by the fact that *joy* is the only positively connotated emotion in the base emotions mix, therefore showing the clearest distinction potential, whereas *anger*, *fear* and *joy* are expected to be more closely intertwined due to their primarily negative connotation.

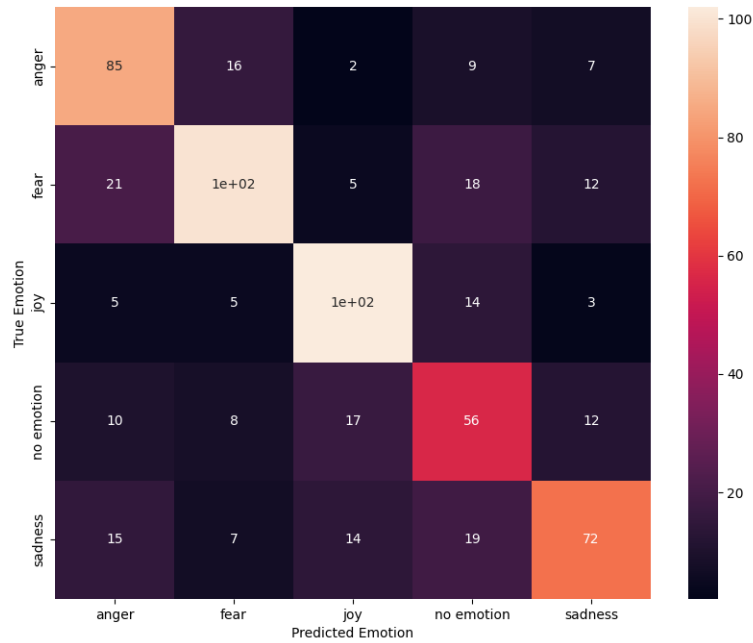


Figure 18. Heatmap for basic emotion performance on test set

This hypothesis is supported by the confusion matrix depicted in figure 18, demonstrating that for the emotion *joy*, the classifier rarely misclassified as one of the three negative emotions (13 cases in sum) and most commonly mistook *joy* for the *no emotion* class (14 cases). In contrast, all three negative emotions respectively show the highest confusion potential for the two remaining negative emotions and the *no emotion* class whilst barely misclassifying as *joy*.

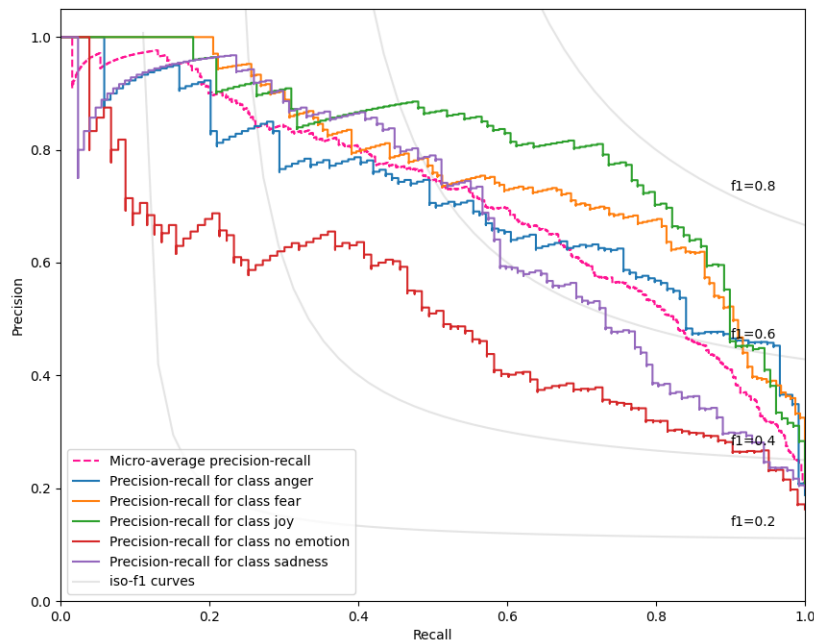


Figure 19. Precision Recall curve for basic emotion performance on test set

Evaluation on test set

Analyzing the complete emotion set, a significant performance difference between the included base emotions and the incrementally included emotions is observed. For the base emotions, F1 scores vary between 0.38 and 0.49 (see table 12), with a large performance drop for the other emotions that yield resulting F1 scores in the range of 0.13 to 0.31. This is underpinned visually in figure 21, with all base emotions plotted above the pink-dotted average line and all supplementary emotions below. Considering the information provided by the Confusion Matrix in figure 20, the inclusion of further emotion categories seems to introduce considerable amount of noise and hardens class segregation.

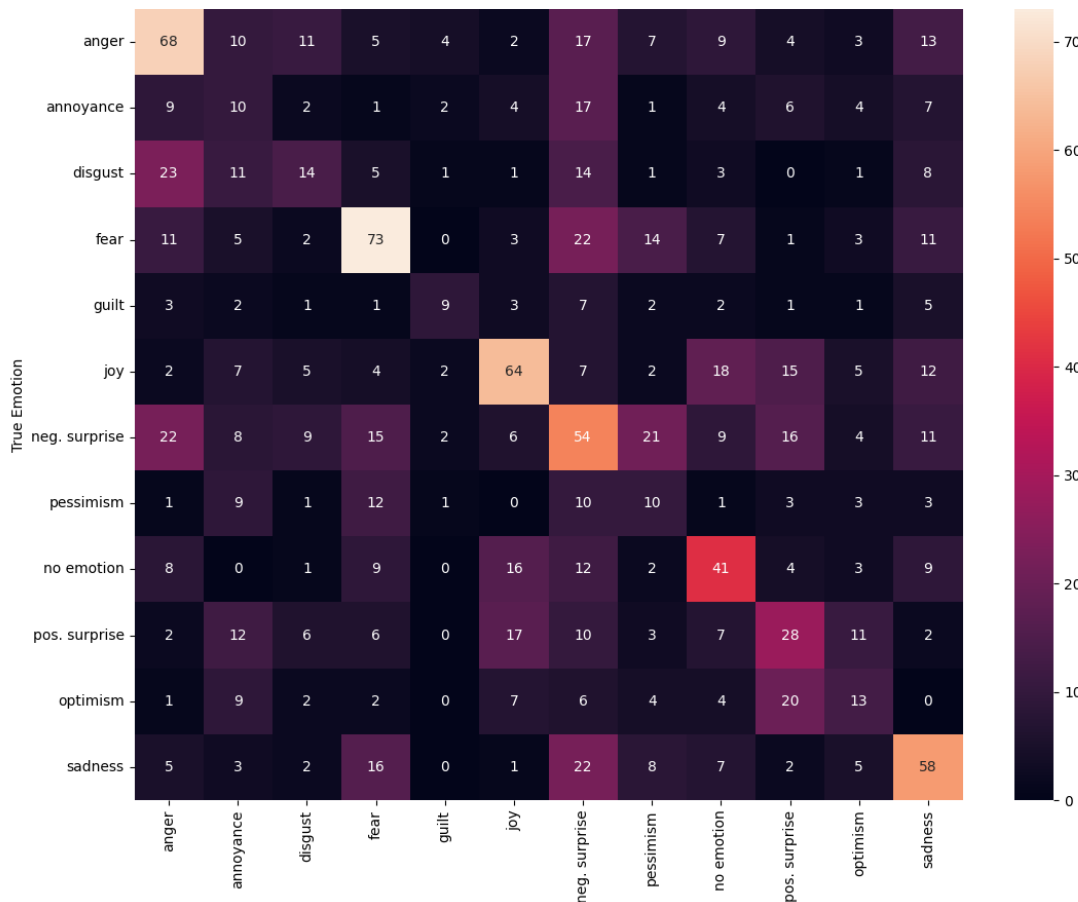


Figure 20. Heatmap for complete emotion performance on test set

Examining the base emotions, their classification performance decreases due to a higher confusion potential with adjacent emotion categories (e.g. *joy* is additionally misclassified as *positive surprise*, *anger* is additionally mistaken for *annoyance*, *disgust* and *negative surprise*). On the other hand, the classifier seems unable to capture and represent the newly introduced emotions properly. For most emotions, the actual class is not the most frequent prediction, which results in an overall poor performance (e.g. *annoyance* is predicted correctly ten times and mistaken 17 times for *negative surprise*).

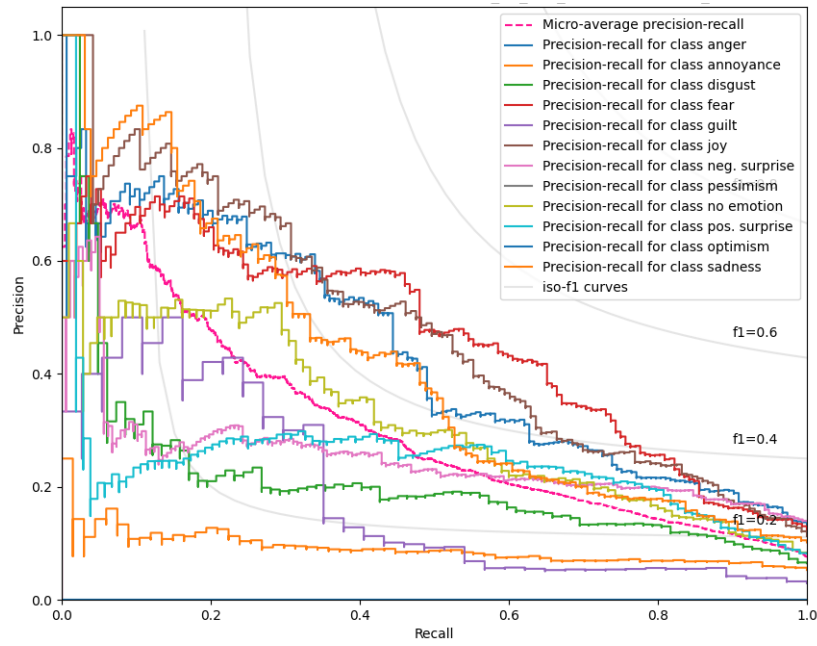


Figure 21. Precision Recall curve for complete emotion performance on test set

In conclusion, the provided results are a clear indication towards the usage of base emotions and support their theoretical foundation.

4.5. Transfer to main corpus

4.5.1. Descriptives for contexts

As a final step, the classifier is transferred to the main corpus that consists of about 4500 real context articles used for advertisement. Tables 13, 14 and figures 22 (left), 23 (left) show the distribution of the resulting emotion classification. For the main corpus, a high imbalance of class prediction is observed. *No emotion* is predicted for most of the cases for both base (74,7%) and complete (59,4%) sets. For base emotions, *joy* is detected in 15,3% of the articles, with the remaining 10% distributed over *anger*, *fear* and *sadness*. For the complete set, *joy* is also second most common with 14,3%. *Sadness*, *anger*, *positive surprise*, *negative surprise* and *fear* are all detected in approximate 5% of cases whereas *annoyance*, *disgust*, *pessimism* and *guilt* form the bottom group that is rarely predicted at all. Notably, the base emotions (and surprise) are detected most often for the complete setting, whilst the additional emotions are rarely represented.

Transfer to main corpus

Table 13

Class distribution for basic emotions on main corpus

Emotion	<i>N</i> contexts	<i>N</i> impressions
Anger	206	48 420
Fear	111	17 416
Joy	695	198 444
Sadness	133	12 732
No emotion	3 383	382 308
Total	4 528	659 320

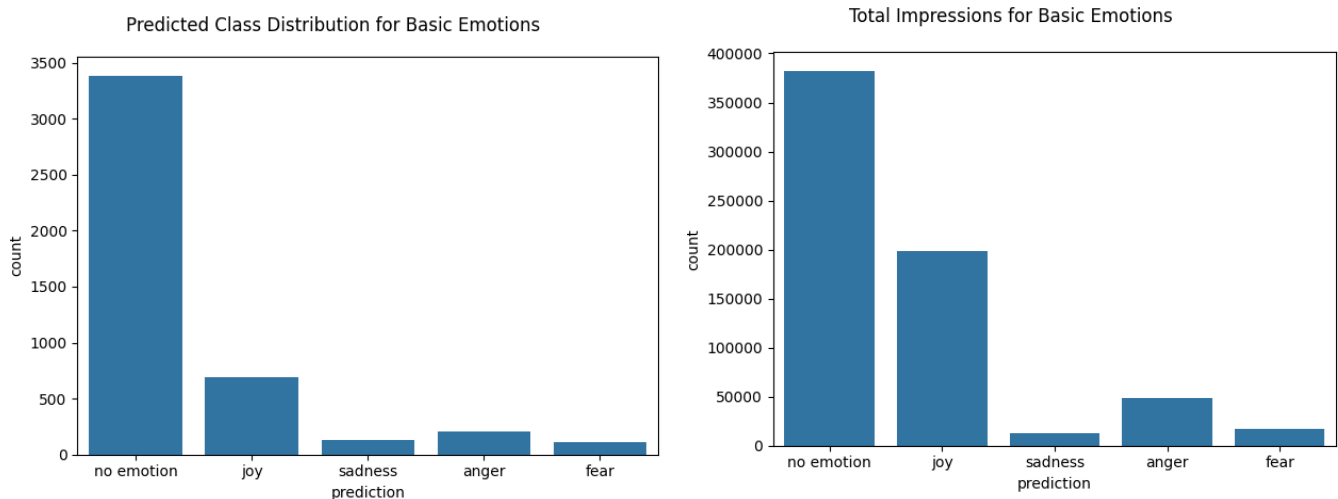


Figure 22. Predicted class (left) and impression (right) distribution for base emotions

Inspecting figure 22 for base and figure 23 for complete emotions respectively demonstrates the divergence between number of contexts and number of impressions. This is caused by the greatly varying number of impressions each article achieves. On average, the used context articles reach 146 measured impressions with a standard deviation of 2336 impressions, with a total range spanning from 4 to 116740 measured impressions.

Transfer to main corpus

Table 14

Class distribution for complete emotions on main corpus

Emotion	N contexts	N impressions
Anger	235	23 488
Fear	222	28 204
Joy	646	84 412
Sadness	223	12 796
No emotion	2 690	329 280
Pos. surprise	202	134 856
Optimism	32	29 504
Neg. surprise	243	16 436
Annoyance	19	180
Disgust	11	128
Guilt	4	32
Pessimism	1	4
Total	4 528	659 320

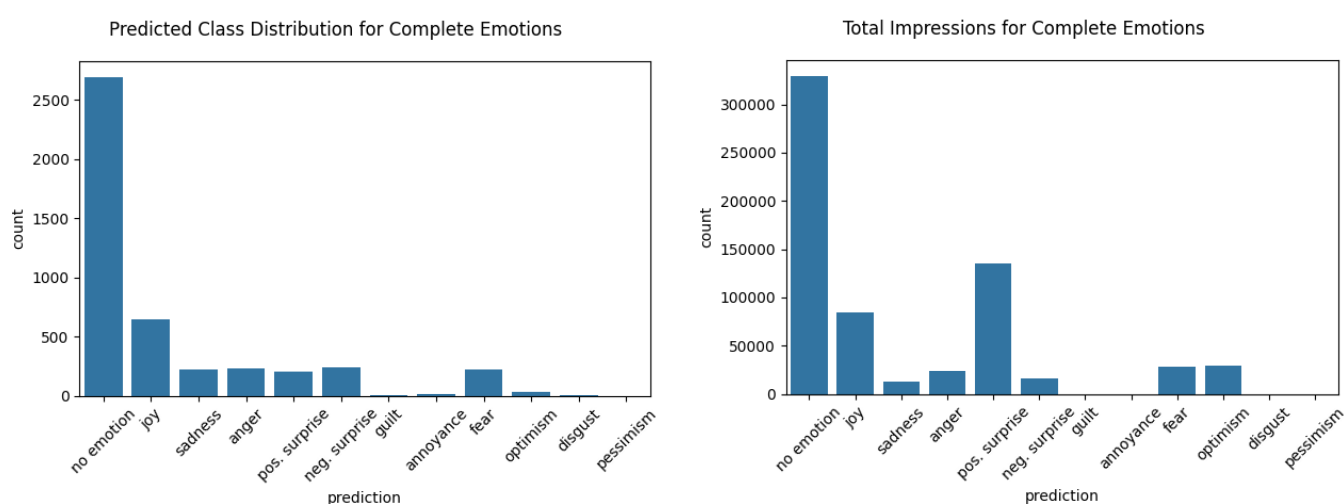


Figure 23. Predicted class (left) and impression (right) distribution for complete emotions

4.5.2. Exemplary evaluation of prediction quality

Since the main corpus consists of unlabelled data, a proper and quantitative evaluation of prediction quality is not possible. As an alternative, a qualitative evaluation is provided focusing on few examples that demonstrate specific qualities and problems regarding generalization capacities on the main corpus. To provide a compact evaluation, the ensuing examples are limited to the four base emotions.

Fear. For the class *fear*, the following examples are chosen:

- (1) *“Vorsicht, die Klausuren kommen: Diese Termine fürs Klausuren Schreiben sollten Sie sich merken”*
- (2) *“Klimaforscher sicher: 1,5 Grad-Limit für Jahrzehnte überschritten”*
- (3) *“Warum sich Chinas Prostituierte vor Kondomen fürchten”*

Examples 1 & 2 are positive examples and show the model’s capacity to consider contextual information instead of relying on mere emotional clue words. Example one evokes fear by warning the reader about upcoming exams, which can be seen as threatening. However, the connection between exams and fear must be implicitly recognised from the context. Example two on the other hand requires the model to recognize the fear inducing effect a report on unstoppable global warming may have. Example three illustrates a case of a false positive, which is most likely caused by the strong focus on the fear related, german word present in the sentence (*fürchten*). Furthermore, this sentence is a prime example for the difficulty the classifier must overcome to differentiate between UGC and PCC. As the task is to predict the reader’s emotion (PCC) instead of the emotion the sentence’s actor is experiencing (which is characteristic for UGC), example three should not be categorised as *fear*, as the fact that prostitutes feel fear in connection with condoms is unlikely to provoke an emotional reaction in the reader, or at most slight amusement.

Anger. For *anger*, the following examples are considered:

- (4) *“Fall George Floyd: Ex-Polizist Kueng erneut zu Haftstrafe verurteilt.”*
- (5) *“Wer AfD wählt, gefährdet Jobs – seinen eigenen und die anderer Leute.”*
- (6) *“Beleidigt und überheblich - der Habeck von der schlimmsten Sorte.”*

Sentence four demonstrates the model’s capacity to extract context again, connecting the George Floyd incident to the viral Black Lives Matter movement and the inherent outrage and anger directed towards police brutality and injustice against people of colour. Examples five

Transfer to main corpus

and six are further examples for arising problems due to PCC. The examples are not considered false positive like example three, as they potentially evoke anger in the reader. The problem lies in the political ambiguity of both sentences, showing clear political orientation the author expresses (UGC). Even though the political message might be persuasive and transmit the desired emotion, the headline's emotionality is very likely to be mediated by the reader's interest in politics and general political orientation.

Joy. Exemplary representatives for joy are:

(7) *“Niederrheinpokal: Burscheider Jugendteam gelingt fast die Sensation.”*

(8) *“Lkw-Fahrer fährt Aktivisten der “Letzten Generation” an – und bekommt Spenden und Jobangebote”*

(9) *“Fränkische Schäuferle mit Biersoße”*

Whilst example seven is a positive example of a correct classification, sentence eight is once again representative for political ambiguity when it comes to emotion elicitation. While some users may feel joy at the news that a truck driver has received job offers and donations as a reward for assaulting a climate protester, this headline is likely to have the opposite effect on others. Example nine is another example of a likely misclassification that is very common for the main corpus. Most food related articles are predicted as joyful content, which might hold true for some, but a classification into the *no emotion* category seems more reasonable in general.

Sadness. For the classification of *sadness*, the following examples are chosen:

(10) *“Flüchtlingsboote kentern: Mindestens 27 Menschen sterben in der Ägäis”*

(11) *“Dieser zum Tode verurteilte Häftling hat kein Recht darauf, schmerzfrei zu sterben”*

(12) *“Krankenschwester gab tödliches Schmerzpflaster - Bewährungsstrafe”*

Sadness classification appears to work best when reporting on individual tragedies or injustices as illustrated by examples ten and twelve. Misconceptions related to tragedies such as death occur mostly when emotional clue words are present as example twelve shows. In this case, a nurse committing crimes that result in a patient's death might cause *sadness* but is more likely to evoke *anger*.

Transfer to main corpus

In summary, the segment demonstrated some characteristics and problems that arise during generalization from training data to main corpus. The following aspects are detected:

- UGC and PCC are hard to differentiate, due to (political) ambiguities (examples 5,6,8) caused by individual preference and ideology, as well as mistaking reader emotions for the protagonists (example 3).
- Especially for the three negatively connotated emotions, the confusion potential and ambiguity is found to be high, which is in line with quantitative findings reported in chapter 4.4.
- A tendency of topic matching is observable, where the classifier matches specific contents to a certain emotion. Examples would be the matching of political contents to *anger*, personal tragedy and death to *sadness* and reports of global catastrophes to *fear*.

4.5.3. Consumer behaviour prediction

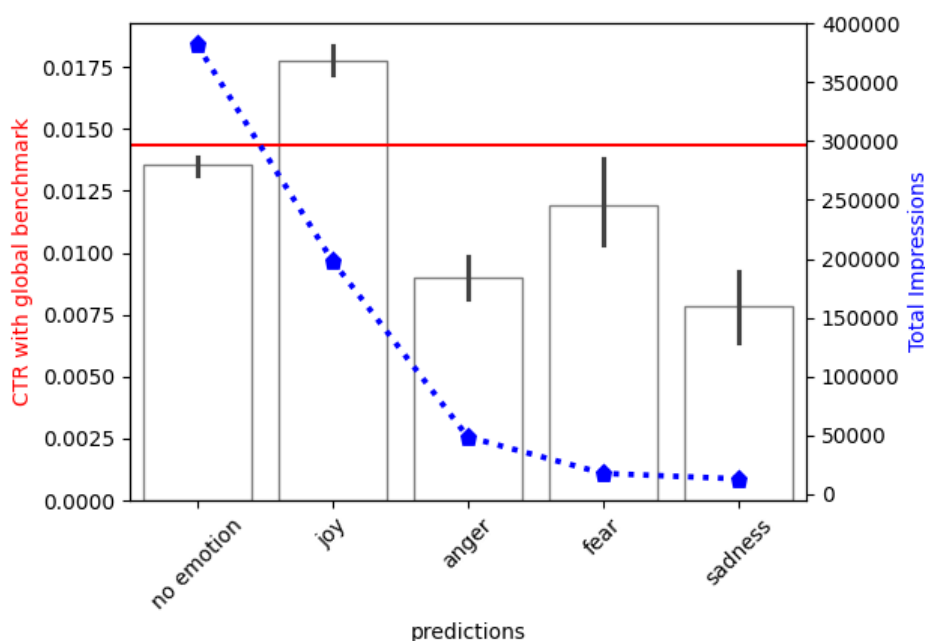


Figure 24. Consumer behaviour prediction capabilities for base emotions

To analyze the actual effect the detected emotion within the contextual article has on customer behaviour, figures 24 & 25 evaluate the CTR on all measured impressions for the classified articles in the main corpus. For both emotion sets, the *no emotion* class contributes the most to the global CTR on the whole dataset, which is expected behaviour as this class is the most frequent one in both cases. Besides the neutral class, the emotion categorization introduces notable variance into the CTR distribution with two aspects that stand out:

Transfer to main corpus

Firstly, for both emotions sets best performances are achieved by positively connotated emotions. For base emotions, *joy* overcomes the global benchmark of 0.0145 significantly by achieving a CTR of approximately 0.0175 whilst the presence of *positive surprise* has an even larger impact on customer behaviour, resulting in a CTR of approximately 0.024.

Secondly, negatively connotated emotions seem to produce lower CTR, with all of them undercutting the global benchmark for both base and complete emotion sets. Amongst those emotions, *fear* is standing out as the only emotion that is close to reach the global benchmark. This can be considered an interesting finding, as it could be interpreted as the proposed flight mechanism *fear* is supposed to evoke. From this perspective, shifting one's attention from the fear inducing context onto the advertisement might be considered a distraction strategy to “flee” from the uncomfortable contents.

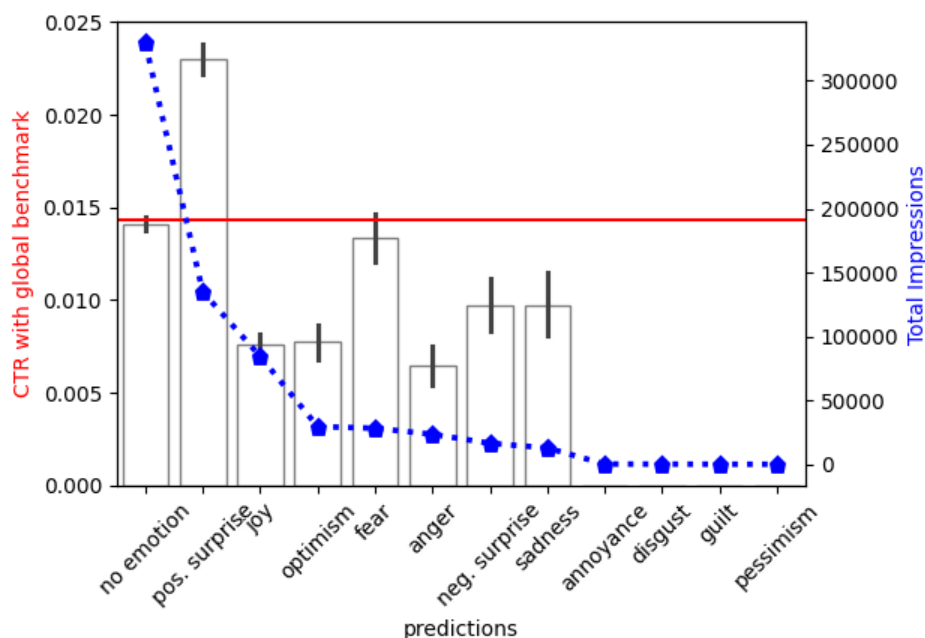


Figure 25. Consumer behaviour prediction capabilities for complete emotions

5. Discussion

5.1. Summary and Conclusion

Scope of this present work was to evaluate approaches to predict customer behaviour in an online marketing setting by solely relying on information inherent to contextual and advertorial contents.

To solve this task, the following steps were taken:

Construct selection to predict user behaviour. Displaying the clearest action tendencies, the construct *emotion* was chosen to predict user behaviour.

Selection of extraction modality & dataset creation. Three labelled emotion datasets were combined that focus on newspaper headlines as the closest approximation for contextual publisher content. Thereby, any information besides textual input as well as advertorial contents were discarded from consideration, reasoned by system requirements and data availability.

Selection of feature sets. ADA, TF-IDF, NRC and linguistic features were extracted from the news headlines and evaluated using several machine learning models.

Training and evaluation of classification algorithms. The experiments revealed that ADA embeddings significantly outperform the other feature sets and yield best results when combined with the NRC features for a NN model architecture. For this final model, it was demonstrated that the four base emotions *fear*, *anger*, *sadness* and *joy* show the clearest differentiation potential, displayed by best classification metrics on class level. This finding supports the outlined theory of base emotions assigning clear and distinct reactional patterns to each base emotion.

Potency of user behaviour prediction on real world data. When applying the resulting classifier onto the main corpus of real publisher articles used in advertisement, a high imbalance of extracted emotionality is detected, with the neutral *no emotion* category being predominantly classified. The analysis of measured consumer interactions with the classified articles of the main corpus revealed that the click rates for presented advertisement is influenced by the detected emotion of the article. Whilst the neutral category closely matched the average click rate, a general trend could be observed, where positively connotated emotions led to click rates above average, while negatively connotated emotions remained below the average value. These findings undermine the importance of context consideration in advertisement and demonstrate

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the manipulative effects that (unconscious) emotional appeal has on human decision and behaviour.

This work contributes distinctly to research on emotion classification in online marketing with the following aspects, as it

- provides an approach to automatically translate and evaluate English text corpora to German to increase dataset size.
- demonstrates the superior performance of GPT-based ADA features for emotion classification compared to other state of the art feature sets and provides first benchmark results.
- contributes to the scarce body of German text classification results and benchmarks.
- provides empirical support for the relevancy of consumer emotionality in online marketing, by comparing advertisement click rates grouped by emotional contents and reporting the resulting click rate variation.

5.2. Limitations and Further Research

5.2.1. Limitations

To interpret this work's results, some limitations need to be considered that are related to model training and model transfer to real-world data.

Training limitations. During model training, handling overfitting was considered the largest challenge that could not be resolved with the help of thorough model tuning. These results reduce the likelihood of model complexity being the problem's cause. Consequently, the overfitting is more likely to be caused by issues related to the data source.

Firstly, even though the present dataset size was significantly increased by translating and combining similar datasets, the resulting dataset size of 6353 samples can be argued to be small compared to other machine learning applications (Demszky et al., 2020; Buechel & Hahn, 2017). The small sample size might hinder the classifier in capturing all relevant aspects to differentiate the target emotions.

Secondly, even though the used datasets are manually labelled following thorough and strict instructions, the question of label quality arises. This problem is most likely not related to lacking quality of the labelling procedure itself but can rather be traced back to the inherent uncertainty and ambiguity of labelling PCC. Being a latent and highly subjective target, labelling a text into an emotion it evokes will clearly produce more noise compared to

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objectively accessible label categories like the detection and distinction of objects in imagery. While there will unlikely be much debate about whether a cat or a horse is depicted in a certain picture, the emotional impact that sentence eight (*"Lkw-Fahrer fährt Aktivisten der "Letzten Generation" an – und bekommt Spenden und Jobangebote."*) has on the reader is highly subjective and is influenced by individual values, experiences and attitudes, inducing a natural source of error and noise to the task of PCC classification. Furthermore, it can be expected that PCC generates more ambiguity than UGC, as example six (*"Beleidigt und überheblich - der Habeck von der schlimmsten Sorte."*) shows. While the emotionality of the author appears obvious to the human reader and seems to be dominated by *anger*, this does not allow an immediate and universal conclusion to be drawn about the reader's emotional response.

This inherent problem is also supported by the authors of the three used datasets (Oberländer et al., 2020; Dang et al., 2021; Strapparava & Mihalcea, 2007), as all of them report rather low inter-rater reliabilities, which is a measure to evaluate the agreement between different annotators that perform the same annotation task. For the *GerSti* ($r = 0.5$) and *AffectiveText2007* ($r = 0.53$) datasets, low average inter-rater reliabilities are reported. Out of the five annotators that labelled the *GoodNewsEveryone* dataset, three of them agreed on the same label in 75 % of all cases. If the criterium is increased to an agreement of at least four annotators, it was only met for 33 % of the cases and only for 16 % of cases all five annotators unanimously chose the same label.

Therefore, the classification accuracy a machine learning application can achieve is limited when human annotators disagree on the labels the model uses for training themselves.

Main corpus limitations. To interpret the reported results, emphasis must be put on the generalization hurdles encountered when applying a classifier trained on manually selected news headline to randomly chosen publisher articles. It is likely that certain aspects are not represented by the rather small dataset used for training that would decrease generalization performance. One such example would be writing stylistics: While the training dataset consists of major newspapers that all are expected to have their stylistic and journalistic standards, the main corpus also includes tabloids that are likely to differ significantly in journalistic style.

Besides generalization problems, the interpretation of CTR evaluation should also be handled with care as it does not account for any confounding variables. This is most prominently demonstrated by the decision made in this work to exclude advertorial content from

consideration, even though it is expected to have interactional effects with emotions detected in its surrounding context.

5.2.2. Further Research

The findings of this work can be considered promising baseline results for further investigation into consumer behaviour prediction for online advertisement:

Including more information. As the present work only considers contextual headlines for classification, feeding more content into the classifier is an apparent starting point for further research.

Firstly, integrating the advertorial contents would allow for richer analysis that could provide beneficial for advertisers. Understanding and exploiting reactional schemes between context and advertisement like a fear-relief pattern (contexts evoking fear should be met by advertisements that evoke positive emotions like joy and hope to relieve the experienced fear) could be valuable and allow for conscious engineering of context and advertisement combinations (Poels & Dewitte, 2019).

Secondly, integrating image features into the classifier to train multimodal models might be a consequential next step, as the interplay of text and image can convey additional emotional nuances, as the example of the crying basketball player mentioned in chapter 2.5.3. illustrated.

Improving data sources. As outlined in the limitations, the observed overfitting is possibly caused by data quality issues. To handle data scarcity, the provision of additional German emotion datasets is of need. Addressing the problem of label ambiguity, following a distributional classification approach might mitigate this problem to a certain degree. Instead of predicting one dominant emotion, these approaches learn a probability distribution over all possible emotion categories and therefore capture ambiguities and mixed emotionality. For image classification, Yang & Sun (2017) demonstrated promising results for this distributional learning approach, which could be applied to textual and multimodal models too.

Further evaluation of main corpus. The analysis of the main corpus included in this work provides first insights into the impact of emotion exposure for online advertisements. Delving deeper into the classified main corpus might provide further findings. As a first proposal, the trained classifier should be applied to more contextual publisher articles to replicate or adapt the found reactional patterns based on the detected emotion. Furthermore, the first evaluation revealed the tendency of topic clusters forming within each emotion (*anger* relates to politics,

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fear to global catastrophes, *sadness* to death and individual catastrophes and so on). Further evaluation in the form of proper topic clustering in relation to the detected emotions might be insightful and help to understand underlying mechanisms and aspects that influence emotion detection in text.

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Appendix

Appendix

Appendix A: Hyperparameter Grid SVM

```
{  
  
'estimator__C': [0.001, 0.01, 0.1, 0.5, 1, 15, 20, 50, 100],  
'estimator__kernel': ['linear', 'poly', 'rbf', 'sigmoid']  
}
```

Appendix B: Hyperparameter Grid NN

```
'method': 'bayes',  
'metric': {'goal': 'maximize', 'name': 'val_f1_metric'},  
'parameters': {  
  'hl1_size': {'values': [1024, 512]},  
  'hl2_size': {'values': [1024, 512, 256]},  
  'hl3_size': {'values': [128, 64]},  
  'lr': {'values': [0.0001, 0.0005, 0.001]},  
  'use_third_hidden_layer': {'values': [True, False]},  
  'batch_size': {'values': [32, 64]}  
}
```

Appendix C: Final model configurations NN

Base Emotions:

- Batch size: 64
- Learning rate: 0.001
- Optimizer: adam

HL1: 1566 X 1024 (relu)

HL2: 1024 X 256 (relu)

HL3: 256 X 64 (relu)

HL4: 64 X 5 (softmax)

Complete Emotions:

- Batch size: 64
- Learning rate: 0.0005
- Optimizer: adam

HL1: 1566 X 1024 (relu)

HL2: 1024 X 256 (relu)

HL3: 256 X 12 (softmax)